



Basic pneumatics (I)



Universidad de Oviedo
Área de Mecánica de Fluidos

Escuela Politécnica de Ingeniería (EPI) de Gijón
4º curso – Grado en Ingeniería Mecánica

**FLUID POWER: HYDRAULIC
AND PNEUMATIC TECHNOLOGIES**



Chapter 3

<https://www.innova.uniovi.es/innova/campusvirtual/campusvirtual.php>



Outline.

- 3.1. Compressed air and pneumatics.
- 3.2. Basic pneumatic system.
- 3.3. Compressors.
- 3.4. Air treatment.
- 3.5. Air distribution.
- 3.6. Actuators. Cylinders
- 3.7. Pneumatic valves.
- 3.8. Vacuum.
- 3.9. Bibliography and additional material.



3.1. Compressed air and pneumatics.

- ❑ The **pneumatic** is the scientific discipline that studies the compresses gases behaviour, considering them for different machines in which they are used. It allows the rational use of the different gases and, particularly, the **compressed air**.
- ❑ The word comes from the Greek word **PNEUMA** that means soul, as the air for the Greeks was the highest matter on earth.
- ❑ The **mechanisms** and **automation** are a fundamental need in the modern industries, promote productivity and improve the working conditions.
- ❑ The pneumatic technology allows a flexible change from manual or semi-automatic industry to a totally automation production. At the moment, there are a wide range of application with **automation** in progress with other technologies, as electrical and electronic.



3.1. Compressed air and pneumatics.

- ❑ The compressed air applications are almost **unlimited**. Nowadays, in Europe, they are the **10%** of the total energy consumption in industry.
- ❑ Some examples of the **variety** and **versatility** of the Pneumatic control are:
 - Hydraulic valves operation.
 - Different movements for the tool and machinery operation.
 - Sprays and industrial painting.
 - Placement and movements in tools and machinery handling.
 - Mechanical fabrication processes (surface curving, isolation, etc.)
 - Welding machinery.
 - Pieces placement to glue, weld, join....
 - Presses and industrial cutters.
 - Bottling machines.
 - Materials and components transportation.
 - Pneumatic handling machines.
 - Automatic calibration and verification.
 - Air suction and vacuum machinery.
 - Robotics and mechatronics, etc.



3.1. Compressed air and pneumatics.

ADVANTAGES of the technology.

- ❑ **Availability.** Most of the industries and plants have compressed air supply.
- ❑ **Storage.** If needed, it is easily kept into pressurized tanks.
- ❑ **Simplicity in design and control.** The pneumatic components are easily designed and mounted to produce an automatic installation.
- ❑ **Flexibility in movements.** Allows the possible movements with the actuators: linear and rotating, with a wide range of speeds.
- ❑ **Economy.** The installation costs are relatively low and repairs or replacements are also relatively cheap. Maintenance is cheap and the fluid is free(*).
- ❑ **Reliability.** Long life and high reliability of any system.
- ❑ **Endurance.** Not affected by high temperatures, dust or corrosive atmospheres.
- ❑ **Environmentally friendly and safe.** Air is a clean product and does not promote any overloading or fire risk.



3.1. Compressed air and pneumatics.

ADVANTAGES of the compressed air.

- ❑ **Unlimited.** As everywhere we are surrounded by air, the fluid is totally available. This fluid is free.(*).Mind: dry and clean air.
- ❑ **Transport.** It is easily transported in pipes, even for long distances. And, in comparison with the oleo-hydraulic systems, the return pipes are not necessary.
- ❑ **Storage.** The compressor does not need to be continuously working. The compressed air might be kept in pressurized tanks.
- ❑ **Temperature resistant.** It keeps its reliability even at extreme temperatures.
- ❑ **Non-explosive or inflammable.** No possible explosion or fire risk.
- ❑ **Clean.** In the case of a system break-down, there is no possible danger for other elements or components. This is really important for the chemical industries, food industry, agriculture applications, textile industry, etc..
- ❑ **Light.** Low inertia, easy control, high speeds, without any over-loading problem, etc..



3.1. Compressed air and pneumatics.

DISADVANTAGES of the compressed air.

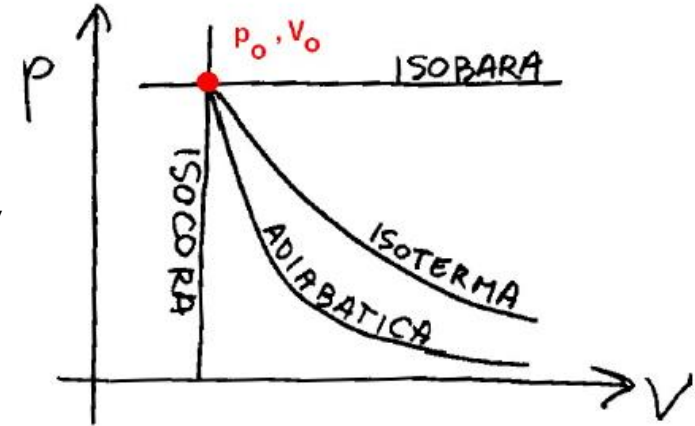
- ❑ **Preparation.** The compressed air needs pre-conditioning before its use. It is mandatory to remove all dust and other particles and humidity to avoid a quick damage of the components (wear).
- ❑ **Compressible.** It becomes somehow difficult to obtain constant velocities in the active elements. At low speed, sudden movements happen, due to the air compressibility.
- ❑ **Force.** The forces in the case of air are limited to **3000 kg**, and the working pressure is often limited to **7 bar**.
- ❑ **Leakages.** When the air has been used, it is discharged to atmosphere, producing a high noise levels, that might be annoying and, even, hazardous. This is somehow overcome using **silencers**.
- ❑ **Cost.** The compressed air is a relatively expensive energy, but this relative high cost is compensated with the good efficiency (repetitive cycles) and the lower cost of the installations.



3.1. Compressed air and pneumatics.

GASES THERMODYNAMIC LAWS.

- The evolution of the ideal gases in **slow processes** (almost steady) is governed by the three basic laws that correlate the volume, pressure and temperature.



- **Boyle-Mariotte Law:** At constant temperature (isothermal process), the relationship between P y V is: $PV = cte$. The gas evolution curves are called then **isothermal**.
- **Gay-Lussac Law:** At constant pressure (isobaric process), the relationship between T y V is: $V/T = cte$. The gas evolution curves are called **isobaric**.
 - For a linear relationship between V and T, it can be written $V = V_0(1 + \alpha T)$ where α is the dilatation gas coefficient at constant pressure (for air, $\alpha = 1/273 = 0.00366$)



3.1. Compressed air and pneumatics.

- **Charles law:** At constant volume (isochoric process), the relationship between P and T is: $P/T = \text{constant}$. The gas evolution curves are called then **isochoric**.

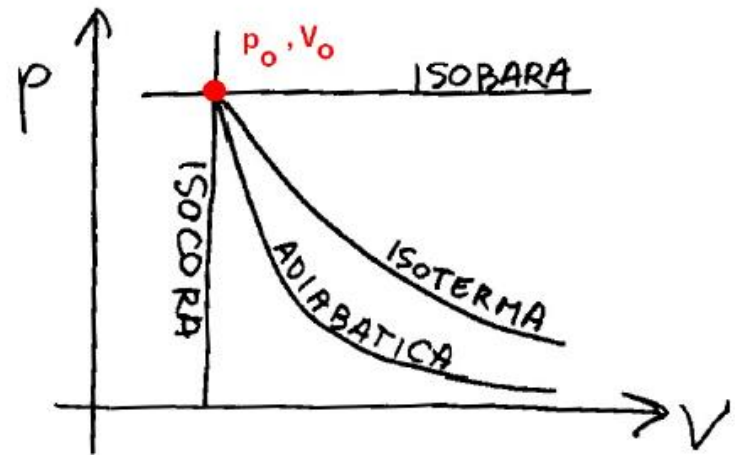
- For a linear relationship between P and T, it can be written as $P = P_0 (1 + \beta T)$ where β is the dilatation gas constant at constant volume (for air, $\beta = 1/273 = 0.00366 \text{ K}^{-1}$).

- **General law:** Combining the previous ones...

$$\frac{PV}{T} = \text{cte}$$

- **Adiabatic process:** When the air evolves **quickly** (very often), there is not enough time to reach equilibrium, and then:

- $n = 1.4$ (Air) $PV^n = \text{cte}$





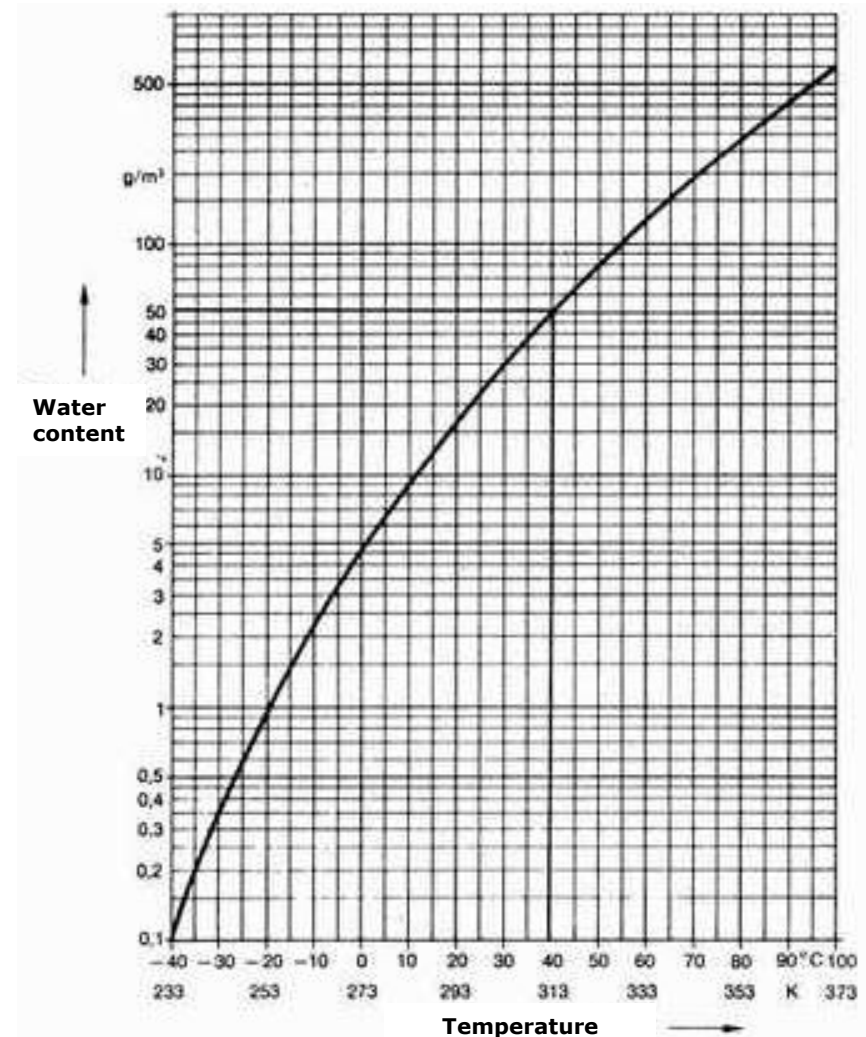
3.1. Compressed air and pneumatics.

Relative humidity.

- ❑ The atmospheric air contains a certain percentage of water (very seldom, it goes to saturation at 100%), that depends on the pressure and temperature.
- ❑ The amount of water in the air (**saturation degree**) **depends almost completely on the temperature** (see graph: amount of water in a m³ of air versus temperature).
- ❑ **Relative humidity:**

$$rh = 100 \frac{\text{Water content}}{\text{Equilibrium content}}$$

Amount of water (gr) in equilibrium for m³ of air (atm)





3.1. Compressed air and pneumatics.

- When the air is compressed, its capacity to admit water in vapour is only equivalent to a “reduced volume”. Therefore, except in the case of a temperature rinsing, the water will **condense in liquid form**.
- **Example:** Obtain the amount of condensed water when 10 m³ of atmospheric air at 15°C and with 65% of relative humidity is compressed to 6 bar (manometric pressure). The final temperature is 25°C.

➤ **Solution:** The calculation of the volume once compressed to 7 bar in absolute pressure:

$$V_2 = V_1 \frac{P_1}{P_2} \frac{T_2}{T_1} \approx 10 \left(\frac{1}{7} \right) \frac{(273 + 25)}{(273 + 15)} = 1.48 \text{ m}^3$$

The amount of water in the initial 10 m³ is obtained from the graph in slide 10. At 15°C, approximately, it is read 12.5 g/m³, and so, a total of 12.5 x 10 = 125 g of water (maximum). As the relative humidity is 65%, then: 125 x 0.65 = 81.2 g.

At 25°C, the saturation graph provides a value of 22.5 g per m³. For the new volume of 1.48 m³, results that only: 22.5 x 1.48 = 33.3 g could be retained. Therefore, the difference would have condensed: 81.2 - 33.3 = **48 g. (water)**



3.1. Compressed air and pneumatics.

Dew point (DP).

- ❑ The dew point determines the **temperature** at which the air reaches the saturation point. This is, when the dry air becomes humid to the maximum that it allows.
- ❑ There would be no condensations if the air temperature is kept over the dew point.
- ❑ For dry air, the dew point is the main parameter that indicates the air dryness degree. Very low dew points indicate a very dry air (good quality air) and high dew points indicate air with high relative humidity.
 - Dry air ($rh < 100\%$), has a dew point always below the atmospheric temperature.
 - For saturated air ($rh = 100\%$), the dew point is at the atmospheric temperature.
 - For saturated air ($rh = 100\%$) but already with liquid in suspension (mist) the dew point is over the atmospheric temperature.



3.1. Compressed air and pneumatics.

- ❑ The dew point can be obtained from the relative humidity and the atmospheric temperature. A diagram with the graph **of the water vapour pressure** is used.
- ❑ Firstly, the humidity for saturation (hs) is obtained, according to:

$$hs = 625 \frac{P_v}{(P - P_v)} \quad (hs \text{ are gr of water for each kg of dry air})$$

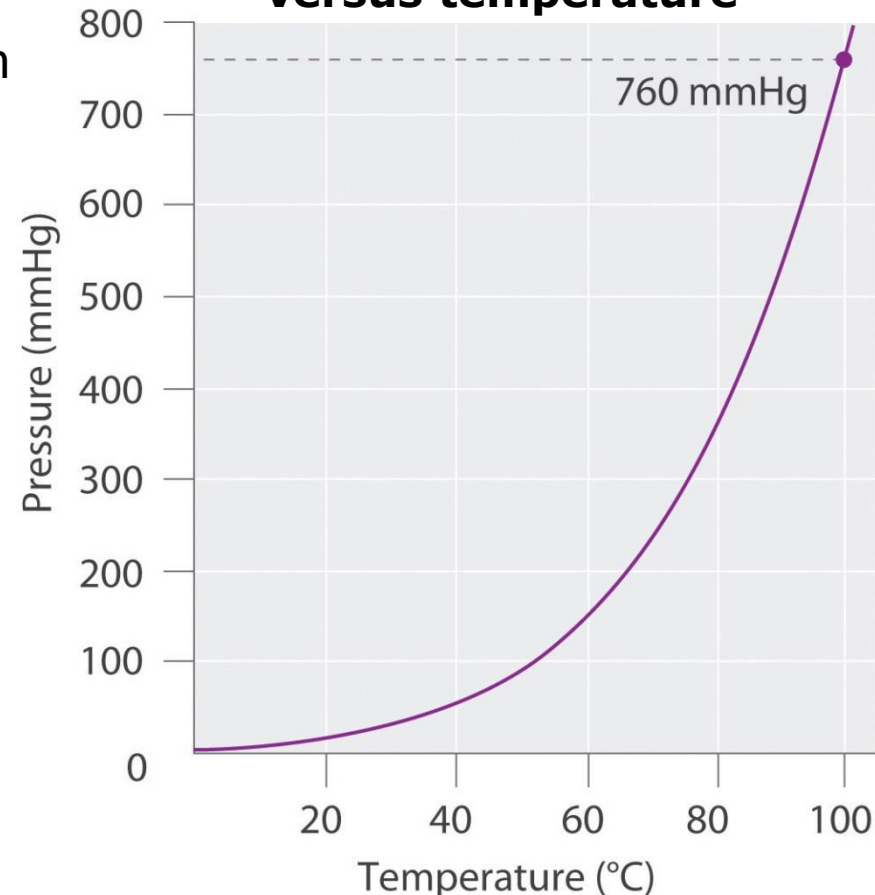
- ❑ Then the absolute humidity is obtained, according to (ah):

$$ah = \frac{rh}{100} hs$$

- ❑ Finally, $ah = hs$ to obtain P_v and the dew point in the graph.

MIND: The value 625 comes from the quotient of the kg of water vapour by kg of dry air, that is: 18 moles of water vapour by 28.8 of dry air: 0.625 kg (times 1000 grams)

Vapour pressure of the water versus temperature



Arden-Buck Eq. (In mmHg for T in °C)

$$P_v = 4.5856 e^{\left(18.678 - \frac{T}{234.5}\right) \left(\frac{T}{257.14 + T}\right)}$$



3.1. Compressed air and pneumatics.

❑ **Example:** Obtain the dew point for a temperature of 30°C and 60% relative humidity at atmospheric pressure.

➤ **Solution:** Firstly, the humidity for saturation is obtained according to:

$$hs = 625 \frac{P_v}{(P - P_v)} = 625 \frac{31.8}{760 - 31.8} = 27.3 \text{ gr}$$

Where $P_v = 31.8$ mmHg is the vapour pressure at 30°C according to the graph and 760 mmHg is the atmospheric pressure. Operating, a 27.3 gr of water for each kg of dry air are obtained at saturation (a similar value would be obtained using the graph in slide 10, but in m³ of air).

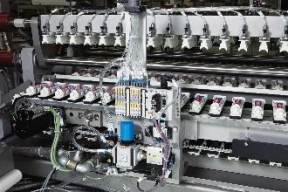
Then, the amount for 60% of relative humidity is obtained:

$$ah = \frac{rh \ hs}{100} = \frac{60 \ 27.3}{100} = 16.4 \text{ gr}$$

These are, then, 16.4 gr of water for each kg of dry air. The vapour pressure is then re-calculated for the obtained amount of water, finding:

$$P_v = \frac{16.4 \ 760}{625 + 16.2} = 19.43 \text{ mmHg}$$

And, then, in the graph of slide 13 again, **the dew temperature would be between 21 and 22°C** (better than using Arden-Buck equation).



3.1. Compressed air and pneumatics.

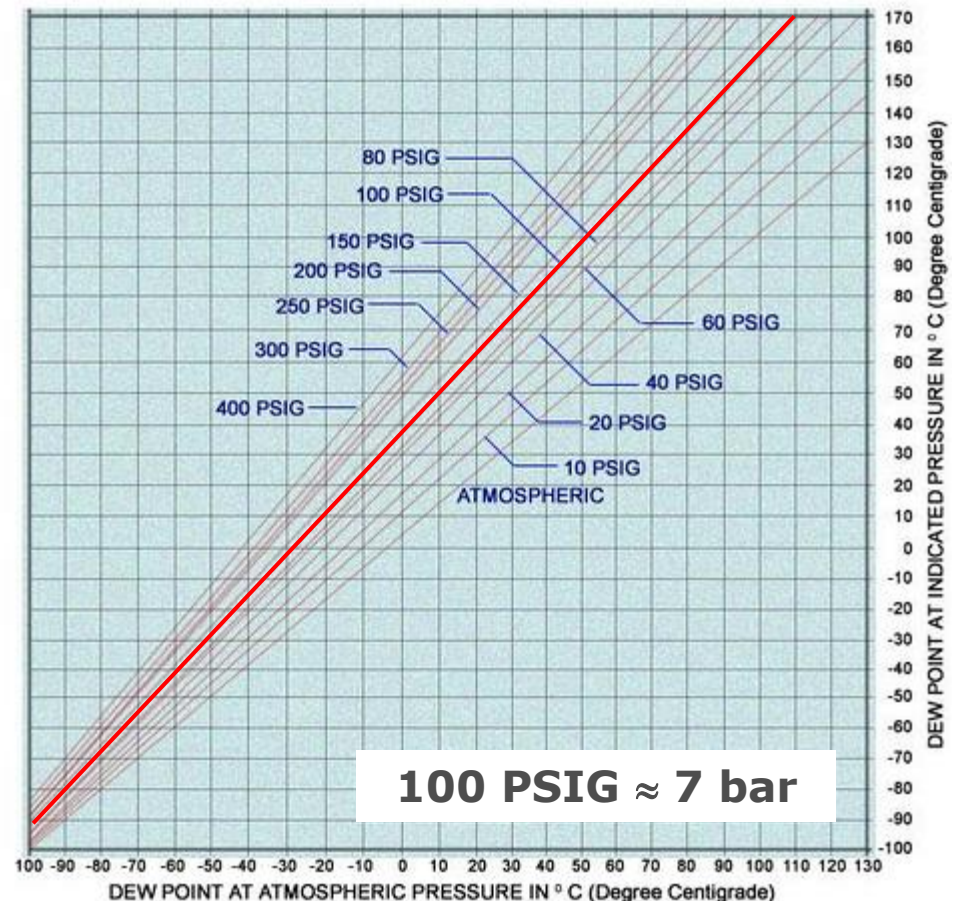
- ❑ **WHEN THE AIR IS NOT AT ATMOSPHERIC PRESSURE,** the before mentioned calculation is done using the graph on the right to correct the dew temperature.

- ❑ **Example:** The calculation already done, but with pressurized air at 7 bar (100 PSIG), would give as dew point:

Using the graph with 21-22°C in the horizontal coordinate and searching for the intersection of the line with 100 PSIG, it is read in the vertical coordinate:

approx. 65°C

Conversion of the dew point under pressure for the dew point at atmospheric pressure





3.1. Compressed air and pneumatics.

❏ **Example 2:** Obtain the saturation humidity at 30°C and a manometric pressure of 7 atm.

➤ **Solution:** Now we have pressurized air, so first we have to obtain the equivalent dew temperature at atmospheric pressure. Using the previous graph, for the saturation, the dew temperature is then 30°C for the vertical coordinate and looking for the line of 100 psig (7 bar, manometric), in the horizontal coordinate, it is read: -4°C. For such temperature, the Arden-Buck equation provides a: $P_v = 3.41 \text{ mm Hg}$.

$$P_v = 4.5856 \cdot e^{\left(18.678 - \frac{(-4)}{234.5}\right) \left(\frac{(-4)}{257.14 + (-4)}\right)} = 3.41 \text{ mm Hg}$$

And, then, at atmospheric pressure, it will be:

$$hs = 625 \frac{P_v}{(P - P_v)} = 625 \frac{3.41}{760 - 3.41} = 2.82 \text{ gr}$$



3.1. Compressed air and pneumatics.

- ❑ Due to the **compressibility** of the air, the relationship between the pressure drop and the flow rate is quite complex.
- ❑ Usually, **there is no need for a very precise calculation** for the pressure and flow rate through the different elements, as the main goal is not this mathematically exact calculation but the automation and correct behaviour of the sequences.
- ❑ Nevertheless, when there is a need to obtain numerical values, several diagrams with flow rate as a function of the cross section of a given element are used. For a **equivalent section (in mm²)** known, different values are obtained if the flow is sonic (**sonic choke**) or subsonic (**un-choked flow**).
- ❑ Those graphs are obtained from the relationships among (P_1 and P_2 , that is the inlet and outlet pressure, in bar and Q in l/min for S in mm²)

Subsonic flow

$$si \quad (P_1 + 1.013) \leq 1.896 (P_2 + 1.013)$$

Sonic flow

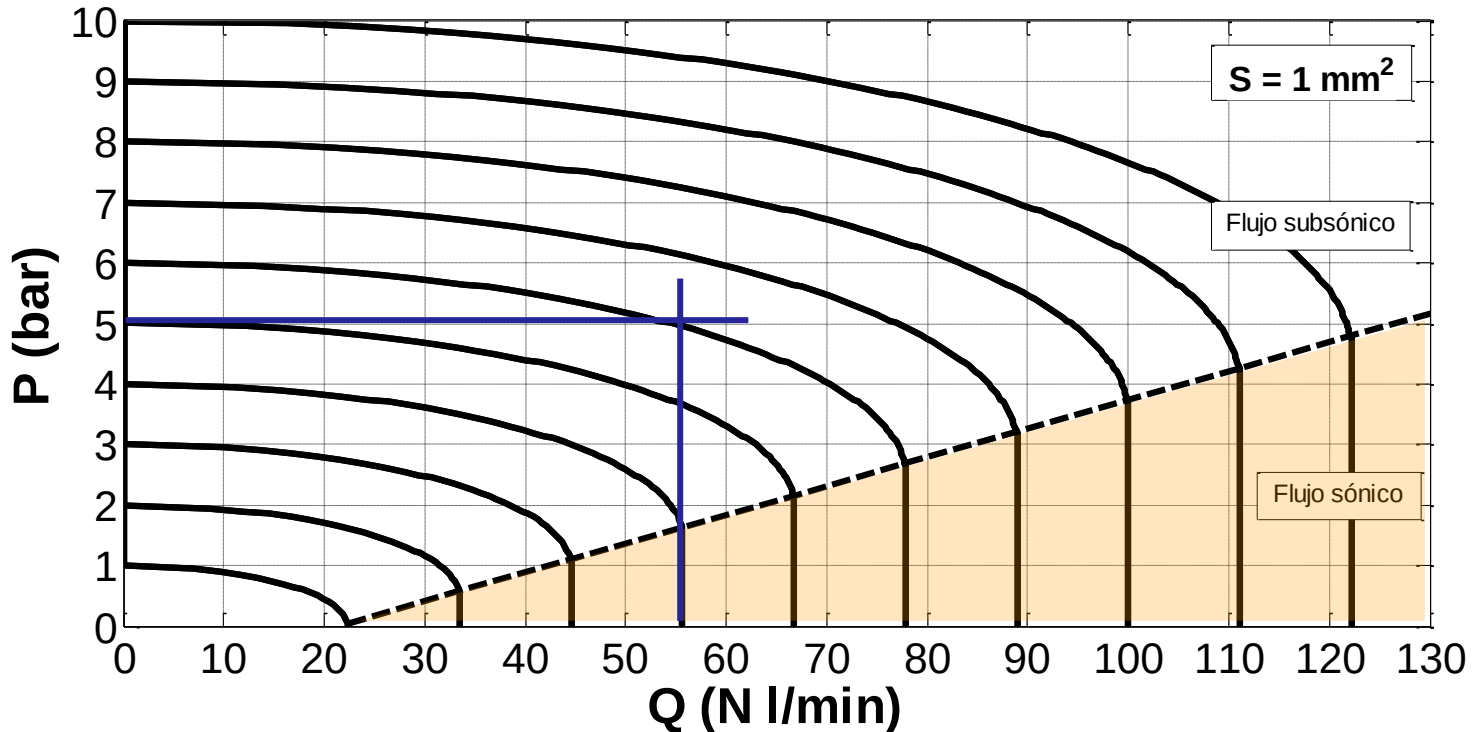
$$si \quad (P_1 + 1.013) > 1.896 (P_2 + 1.013)$$

$$Q = 22.2 \ S \sqrt{(P_2 + 1.013)(P_1 - P_2)}$$

$$Q = 11.1 \ S (P_1 + 1.013)$$



3.1. Compressed air and pneumatics.

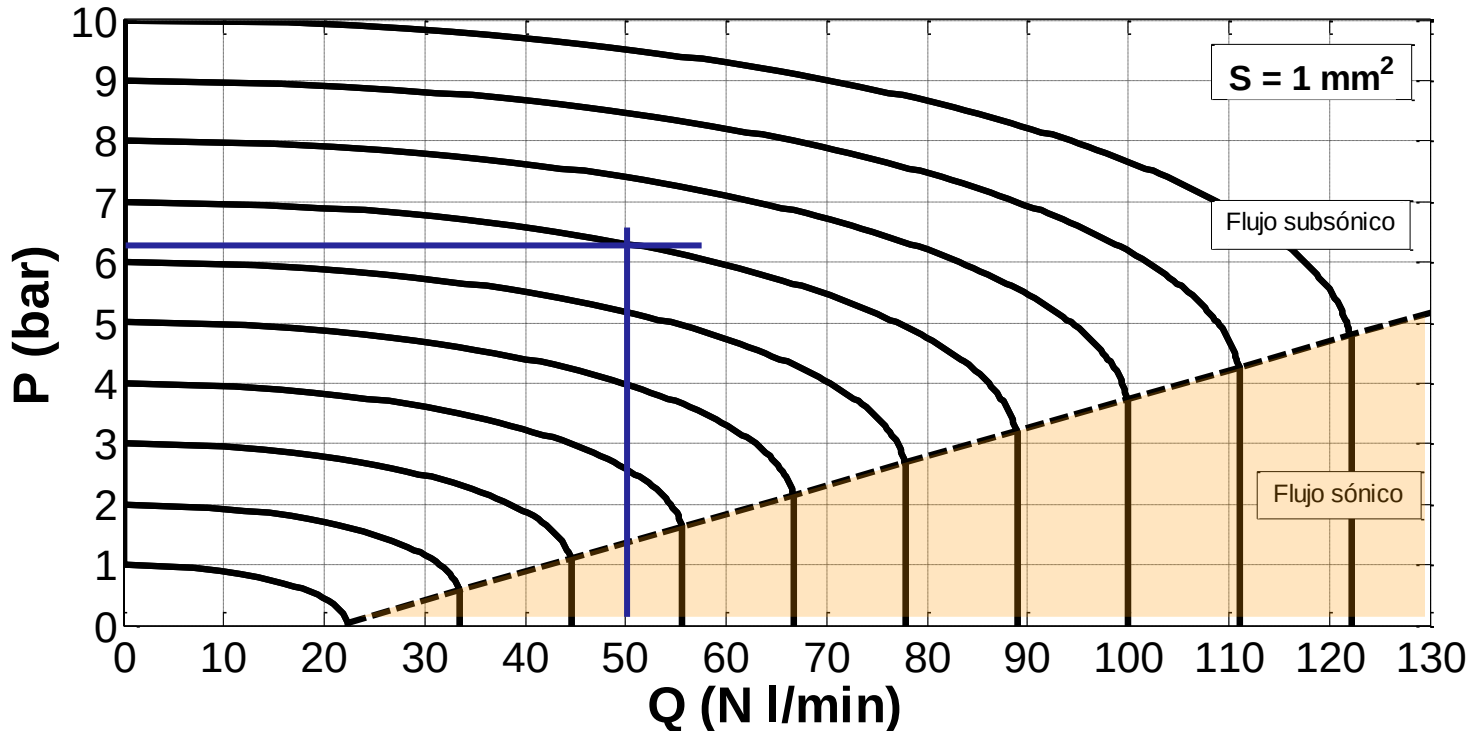


❏ **Example:** A pneumatic element with $S = 4.5 \text{ mm}^2$ has an inlet pressure of 6 bar. If the pressure drop is 1 bar, obtain the flow rate.

➤ **Solution:** Using the curve from 6 bar until it reaches the horizontal line of 5 bar, a value of 55 Nl/min (normal litres by minute) it is found. As the cross section is 4.5 mm^2 , and the graph is for 1 mm^2 , the right value will be obtained multiplying by 4.5, and then, the right value is 245 l/min.



3.1. Compressed air and pneumatics.



❏ **Example 2:** Obtain the pressure drop in an element with $S = 12 \text{ mm}^2$ which has an inlet pressure of 7 bar and a flow rate of 600 Nl/min.

➤ **Solution:** This flow rate for an element of section 1 mm^2 , would be $600/12 = 50 \text{ Nl/min}$. Entering with this value and going till the curve of 7 bar we obtain a crossing point. Then, the horizontal from this point will give the exit pressure (approximate 6.3 bar). Thus, the pressure drop is 0.7 bar.



3.1. Compressed air and pneumatics.

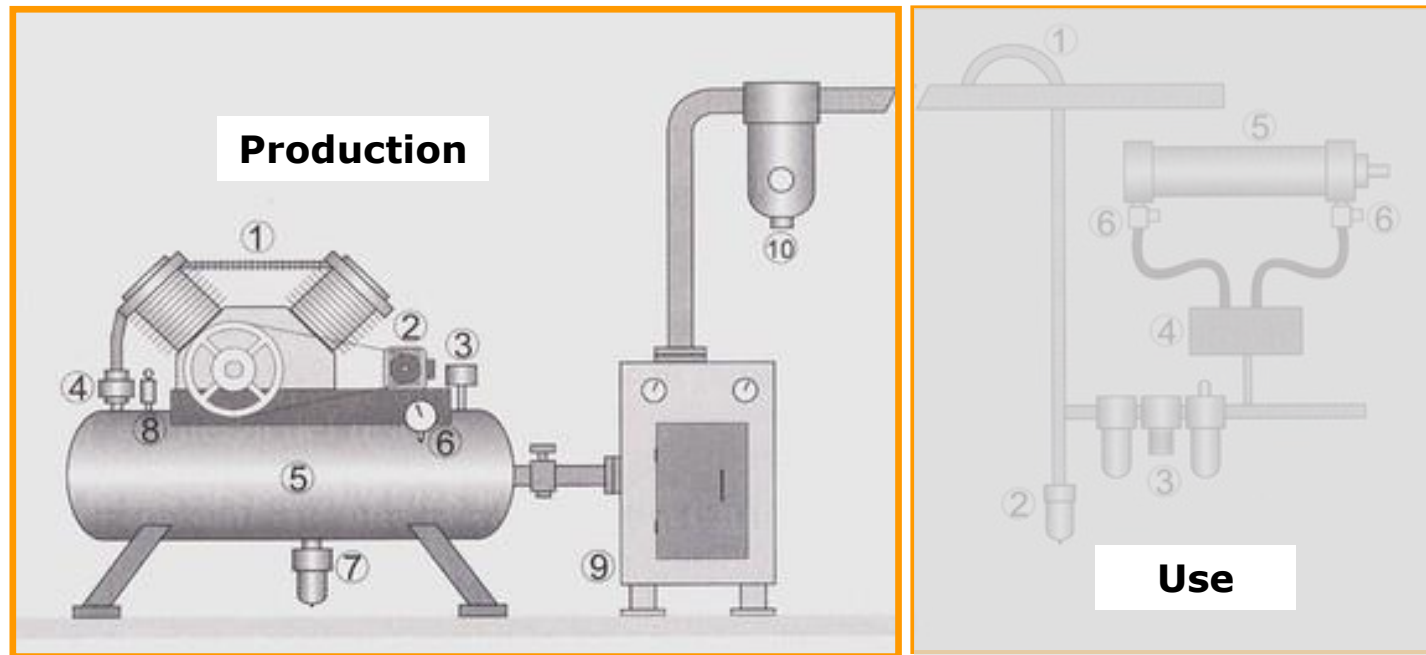
- ❑ Even though the water in a given air volume depends only on the temperature, **a moderate compression at 7 bar reduces not only the air volume** but the amount of water that could be inside the air, and **condensation** may show up. Then, it is mandatory to include **dryers and drainage**.
- ❑ Besides, as the air after the mechanical compression leaves with higher temperature, an additional **cooling** to drop the temperature is needed. In such way, a condensation is induced and then no humidity is allowed in the system (pipes, where it would probably condense latter). Therefore, at the exit of the compressor, **a cooler is placed**.
- ❑ Moreover, additional **dryers** to collect the water from the condensation are needed. The final objective is to introduce **dry air** with high standards.
- ❑ The head losses calculation and the flow rates through the elements is quite complex, but can be approximated by using certain experimental **graphs** that consider the possible sonic choking of the cross sections.



3.2. Basic pneumatic system.

Air production system.

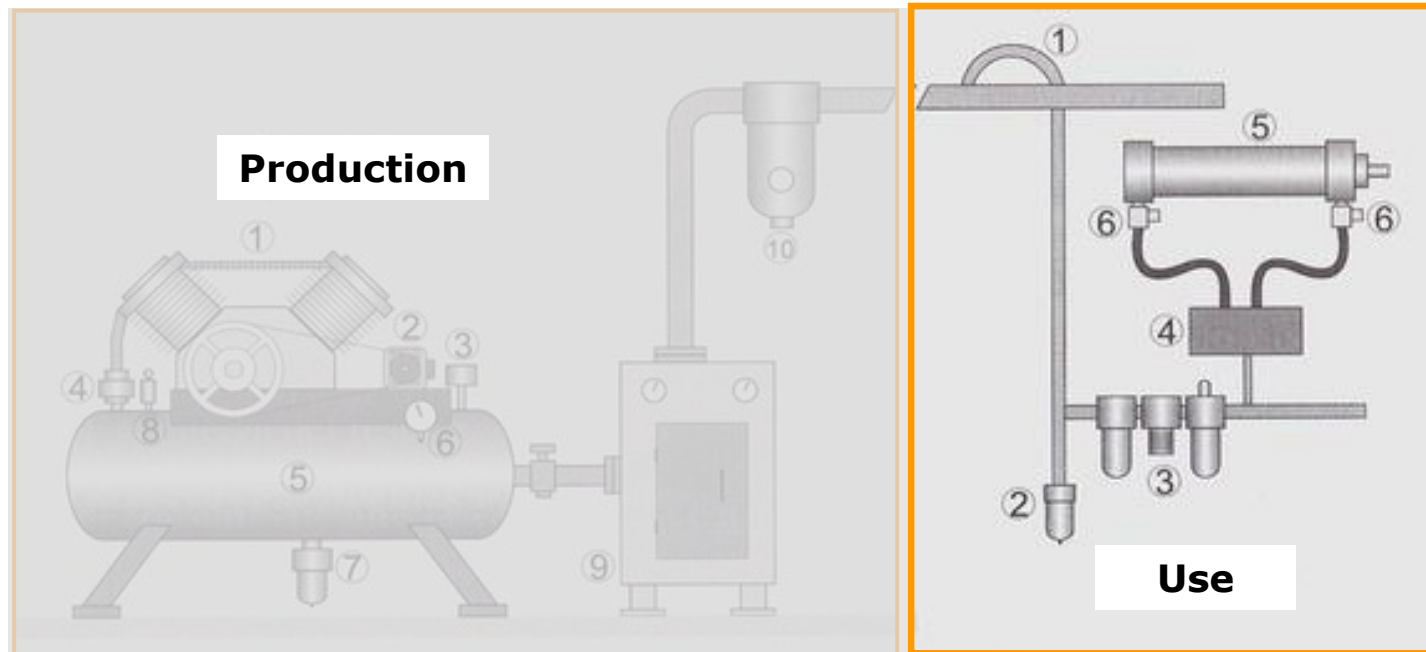
- | | |
|-----------------------------|----------------------------|
| 1) Compressor. | 6) Manometer. |
| 2) Electrical motor. | 7) Automatic drain. |
| 3) Pressure limiting value. | 8) Relief valve. |
| 4) Non-return valve. | 9) Refrigerated air drier. |
| 5) Tank. | 10) In-line filter. |

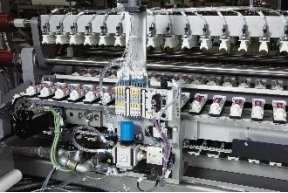




- 1) Air drain with filter.**
- 2) Automatic drain.**
- 3) Air conditioning unit (filter-flow control-lubrication, FCL)**

- 4) **Directional valve.**
- 5) **Actuator.**
- 6) **Flow/speed control.**





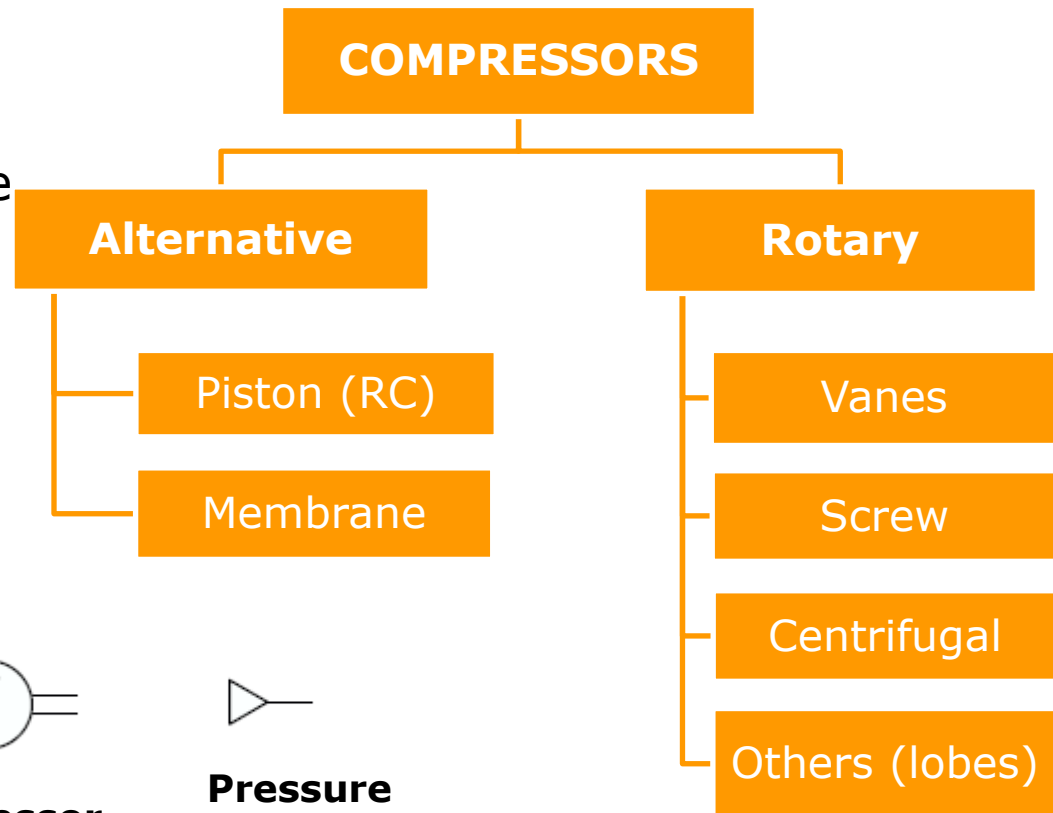
3.3. Compressors.

Definition.

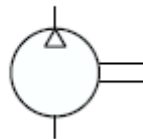
- ❑ A **compressor** transforms the mechanical energy coming from a primary mover (electrical motor or internal combustion engine) into pressure power/energy for the compressed air.

Classification.

- ❑ Two main types:
 - **Alternative** (Positive displacement machines)
 - **Rotating** (both positive displacement machines and turbomachinery)



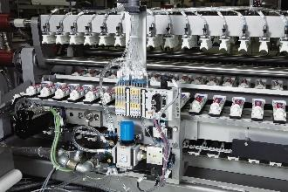
Symbols:



Compressor



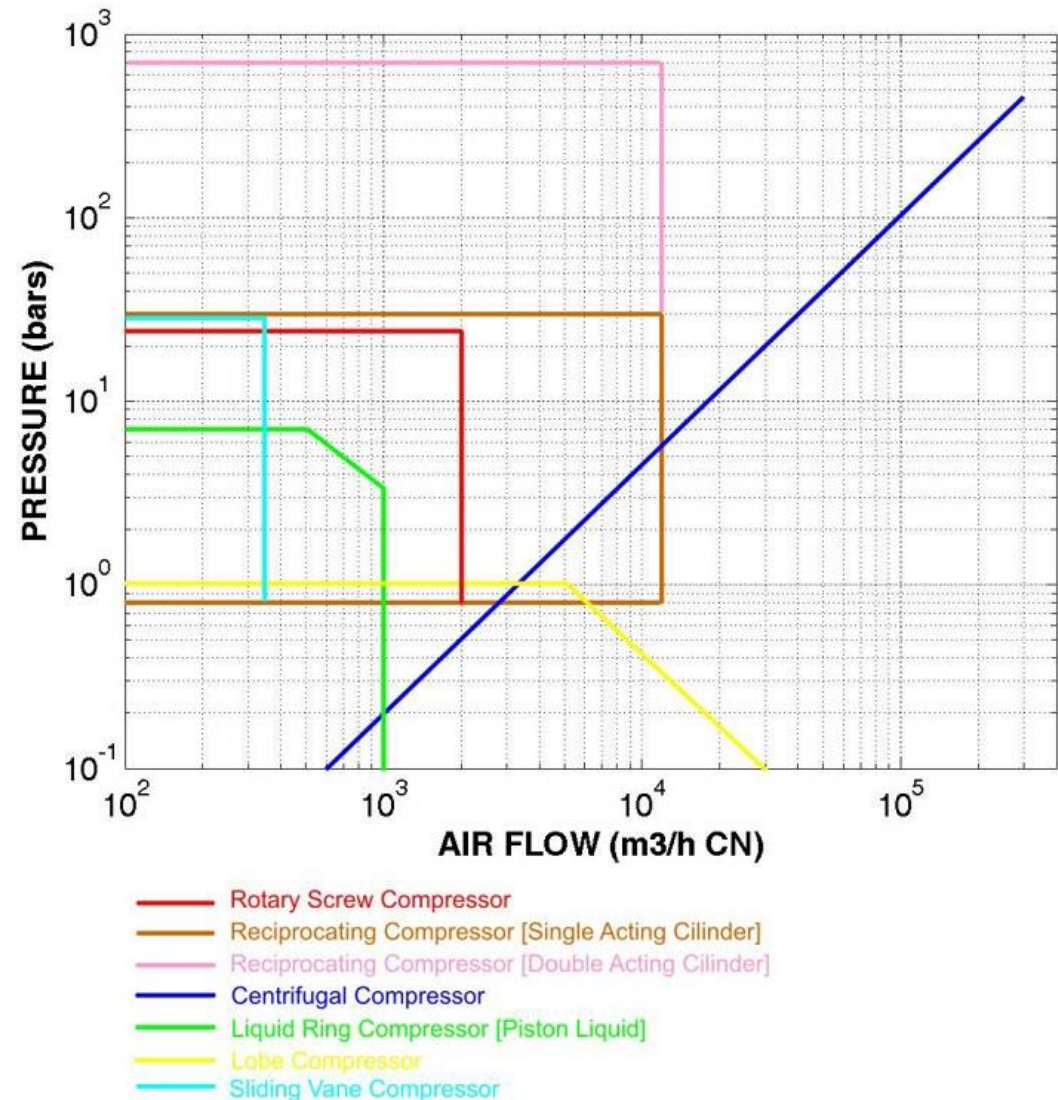
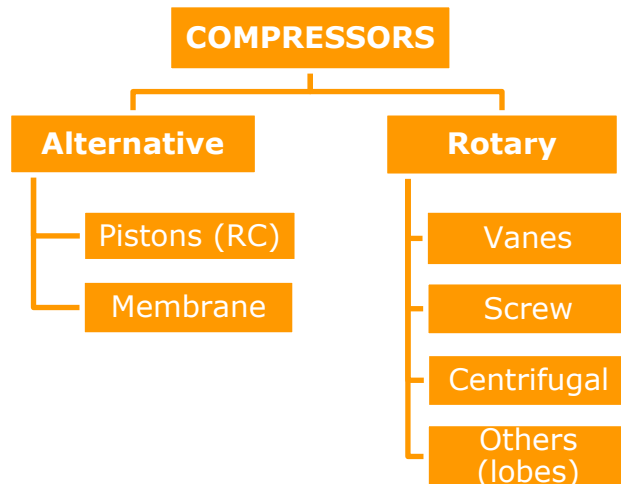
Pressure point



3.3. Compressors.

Selection.

- Depending of the flow rate and pressure range required.

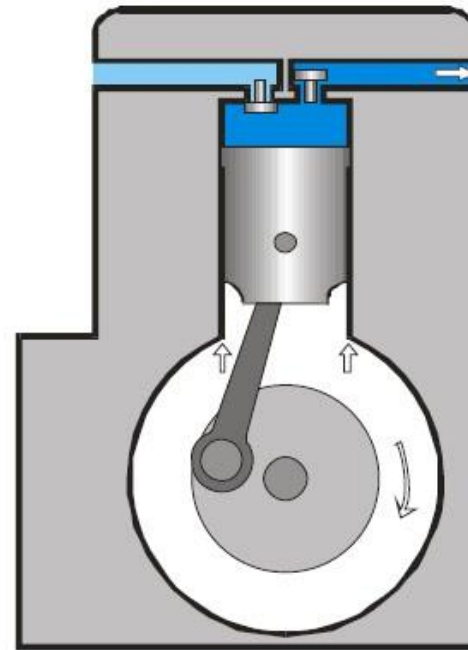
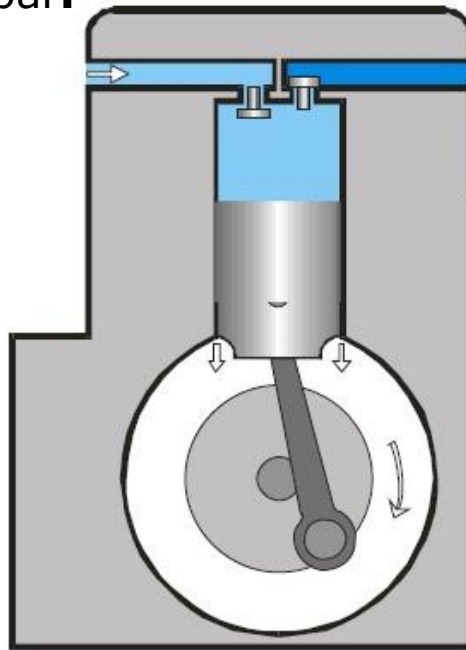




3.3. Compressors.

Plunger, one stage.

- ❑ It is the type of compressor **cheapest and most robust**. On the other hand, it **requires lubrication** for its normal working and promotes a big temperature increase (air taken at P_{atm} and 30°C it is possible to reach 247°C when compressed at 7 bar).
- ❑ It may be used for fixed or mobile equipment, in a wide variety of sizes. The bigger ones can deliver up to $500 \text{ m}^3/\text{min}$. The attainable pressures could reach 6-7 bar.

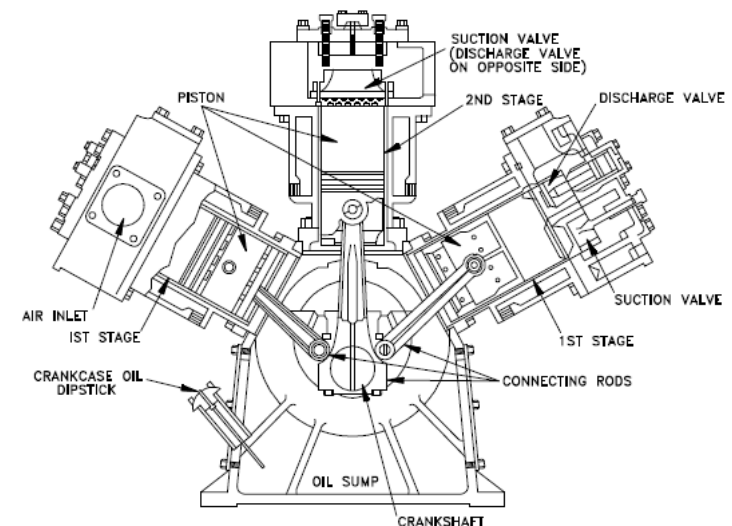
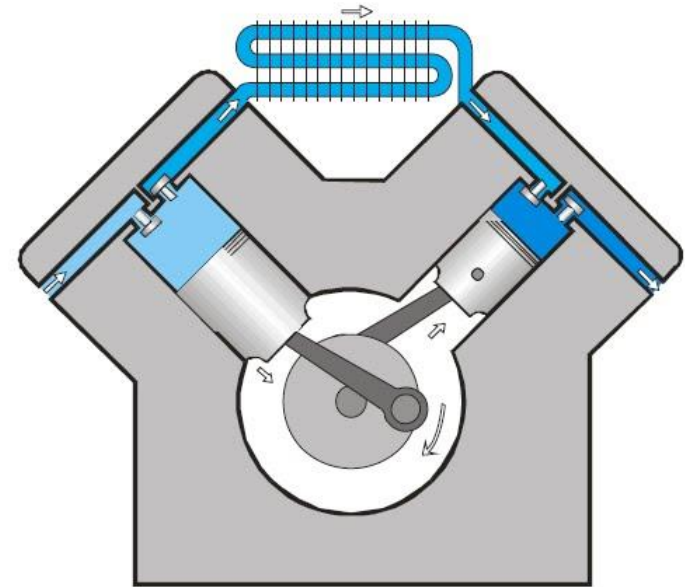




3.3. Compressors.

Plunger, two stages.

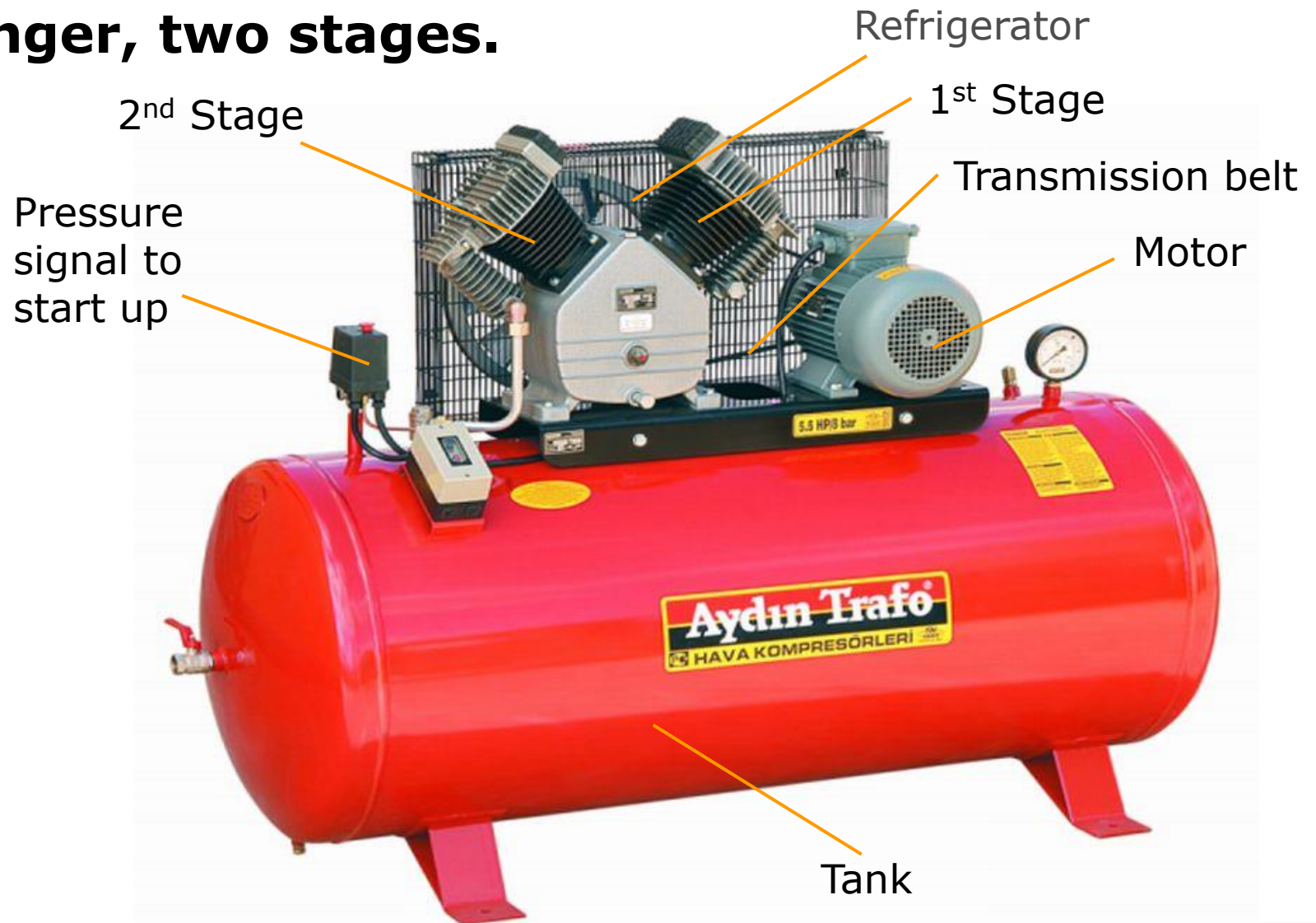
- ❑ When one stage compressors work in values upper than 6 bar, they generate a quite high heating and subsequent **efficiency loss**.
- ❑ Usually, the most part of the pneumatic systems use two stage compressors.
- ❑ The first stage reaches up to 3 bar, then a cooling is performed, and the second piston is able to reach 7 bar.
- ❑ The final temperature could reach **120°C**.
- ❑ Classical with **V cylinders**.





3.3. Compressors.

Plunger, two stages.

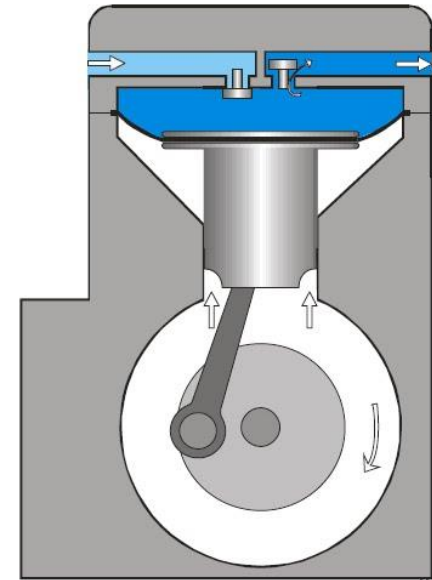




3.3. Compressors.

Membrane or diaphragm.

- They deliver the dry compressed air up to 5 bars and **totally oil-free**: a membrane is between the air and the piston, so it increases the useful surface and avoids that the lubrication oil could get in contact with the air.
- They are used in food industry, pharmacy, etc.
- Usually, they deliver no more than 30 m³/h.
- Also, they are used for pressures under 7 bars.

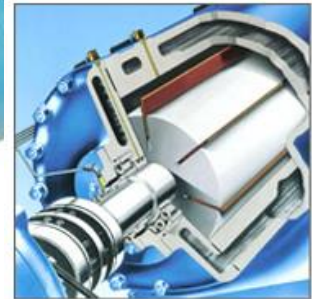
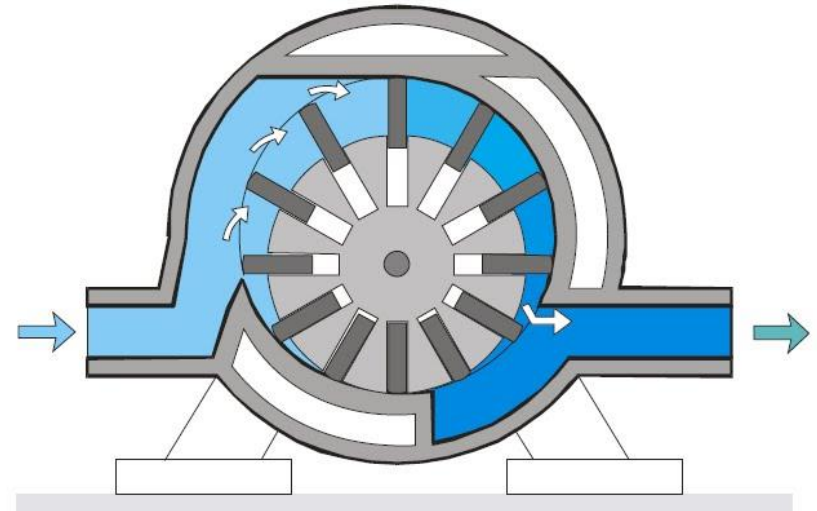




3.3. Compressors.

Sliding vanes.

- ❑ Include a rotor with a certain eccentricity and a series of vanes that move in linear movement in an equally spaced radial slides.
- ❑ They are lubricated by oil at the inlet, and this is also used to refrigerate and keep temperatures under 190°C.
- ❑ They are often used in systems requiring less than 150 m³/h at a maximum pressure of 7 bars.

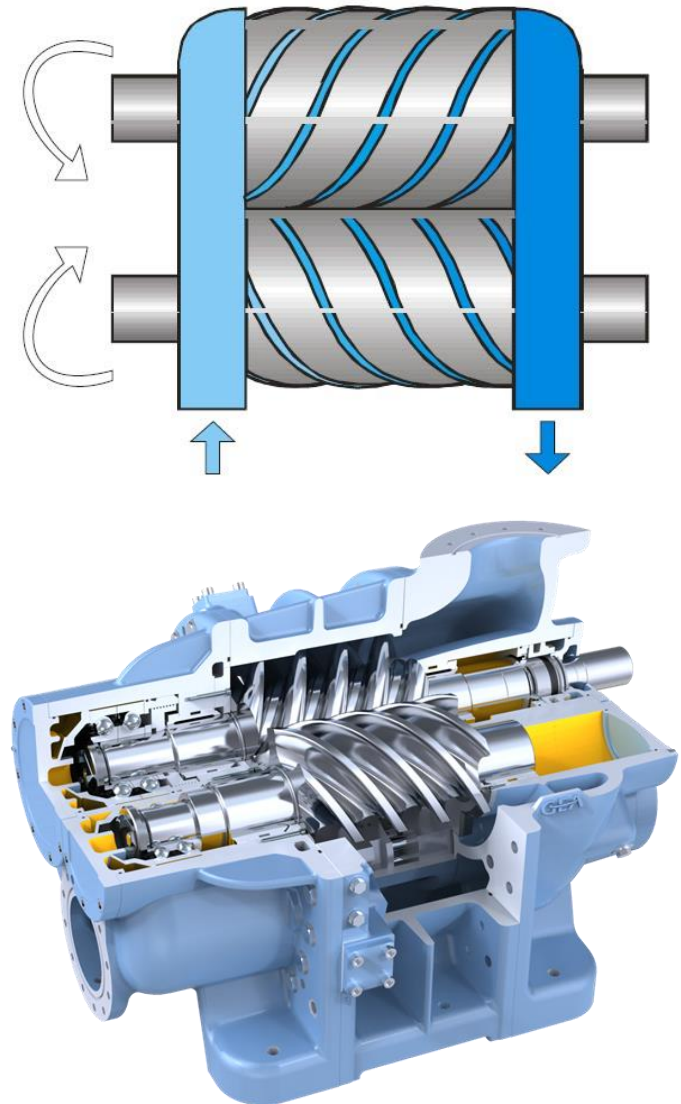




3.3. Compressors.

Screw type.

- ❑ Its use is quite widespread in industry, together with the plunger ones.
- ❑ They work by the rotation of two helicoidally manufactured shafts, joined parallel, and that are inside a metal housing, moving air in a continuous way. The volume between the rotors is being reduced and the air inside compressed.
- ❑ A lubrication is needed for the moving rotors with oil, to avoid severe wear and cool the elements. They can provide high flow rates, around 400 m³/min and pressures around 10 bar. They can also provide different stages, reaching, then up to 30 bar.

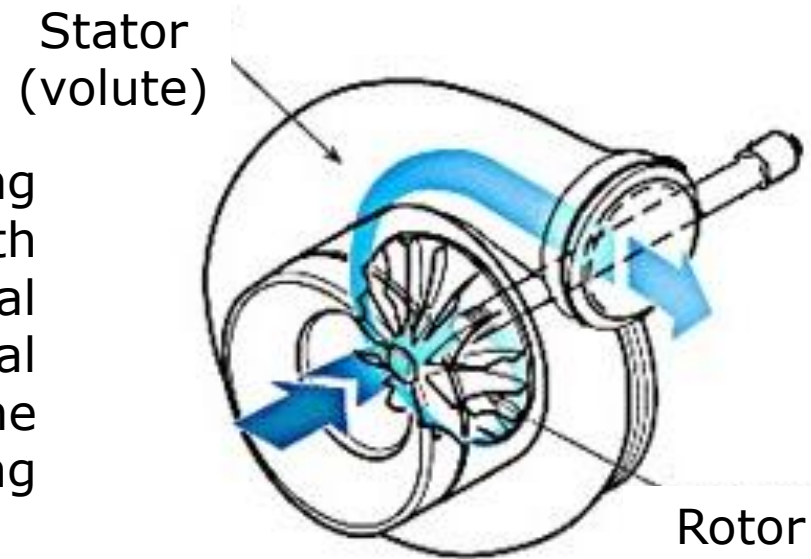


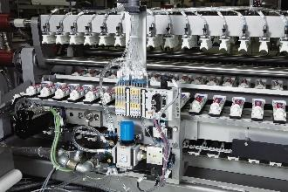


3.3. Compressors.

Centrifugal.

- Based in the compression of air by using the centrifugal force and are built with an impeller that rotates inside an spiral casing (volute) and taking air in an axial direction, deliver it in a radial one. The centrifugal force acts on the air, rising its energetic level and compressing it.
- They can work as single or multiple stages, reaching pressure levels of around 8-12 bar and flow rates up to 10000 - 20000 m³/h. The rotating speeds are quite high, around 15000 - 20000 rpm.





3.3. Compressors.

Cubic capacity of a compressor.

- ❑ The **delivered flow rate** from a compressor is measured as standard atmospheric air or flow rate and it is expressed in Nm^3/min or NI/min (**standard conditions at 20°C and P_{atm}**). It is calculated as the cubic capacity of the each cylinder by the number of cylinders in the first stage and the rotating speed.
- ❑ If a two stage compressor is being studied, only the cylinders in the **first stage** are considered.
- ❑ As in any machine, the **volumetric efficiency** can be obtained as the quotient between the real flow rate and the theoretical one.
- ❑ Up to compression ratios of 4/6, the efficiency drops and, then, it would be more convenient to place 2 or 3 stages with lower ratio:

(p_1/p_2)	$\eta_{\text{vol}} \%$	$\eta_i \%$	$\eta_{\text{mec}} \%$	$\eta_{\text{eléc}} \%$
2	88	90	85 a 93	85 a 90
4	83	82	85 a 93	85 a 90
6	78	74	85 a 93	85 a 90



3.3. Compressors.

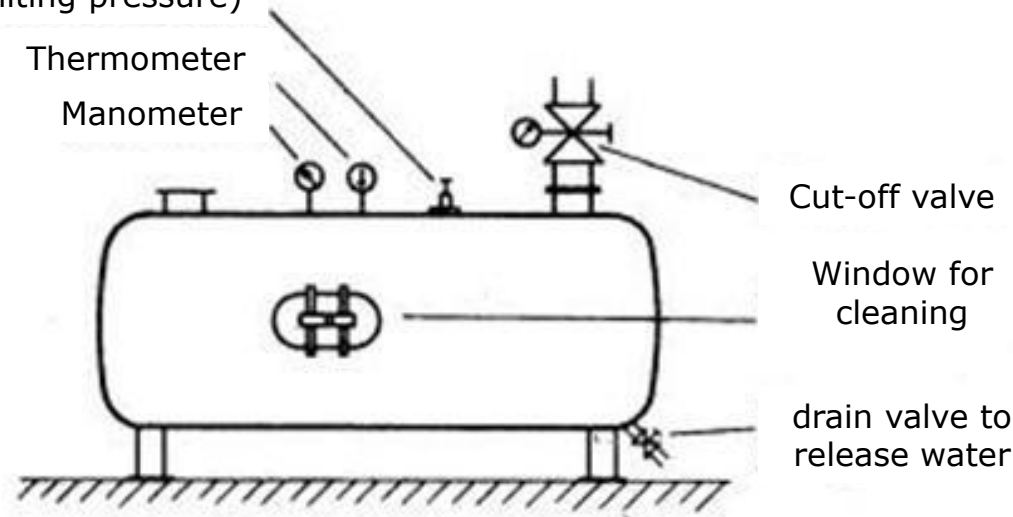
Tank. Definition and elements.

- ❑ A tank acts as a **pressurized accumulator** usually built in metal plate, either horizontally or vertically mounted, that keeps the pressurized air and dampens the possible flow rate fluctuations in the pipes.
- ❑ It includes: relief valve, pressure switch with maximum-minimum values to control the compressor, a manometer, optionally a thermometer, cut-off valve, drain valve to eliminate the water and a window for cleaning.

Pressure relief valve (limiting pressure)

Thermometer

Manometer



Cut-off valve

Window for
cleaning

drain valve to
release water



3.3. Compressors.

Tank. Functions.

- ❑ The tanks main goal is to keep air at pressure coming from the compressor.
- ❑ It should compensate the pressure fluctuations in the piping system, which are more common when the compressor is of the plunger type.
- ❑ They allow stoppage times to the compressor, improving the thermal equilibrium and the useful life.
- ❑ They also allow the cooling of the air coming from the compressor. Their size conditions this cooling effect and also, the water contained in the air. It is therefore quite recommended to place them in fresh places of the installation.
- ❑ The cooling implies the condensation of part of the humidity of the air to water. This water is eliminated through the drain (automatic).
- ❑ They also retain dirtiness from the compressor, so it may be considered as a first filter in the system.



3.3. Compressors.

Size selection.

- They often have cylindrical shape with spherical sector ends. Very often their shape is with a length 2 to 3 times their diameter.
- To obtain the volume, an approximate equation can be used:

$$V_0 = \frac{1000 Q_n}{Z (P_{\max} - P_{\min})}$$

- Q_n : delivered flow rate by the tank in m³/h.
 - $P_{\max}$: Maximum allowable pressure in bar.
 - $P_{\min}$: Minimum allowable pressure in bar.
 - Z : Switch on/off the compressor per hour.
 - V_0 : Initial volume of the tank (at atmospheric pressure 1013 mbar).
-
- **Example:** Tank to deliver 30 m³/h, with 5 switching on-off/hour. Pressure range: 100-80 psi (7 - 5.5 bar). **Solution: 4000 litres.**



3.4. Air treatment.

After coolers and Post-coolers.

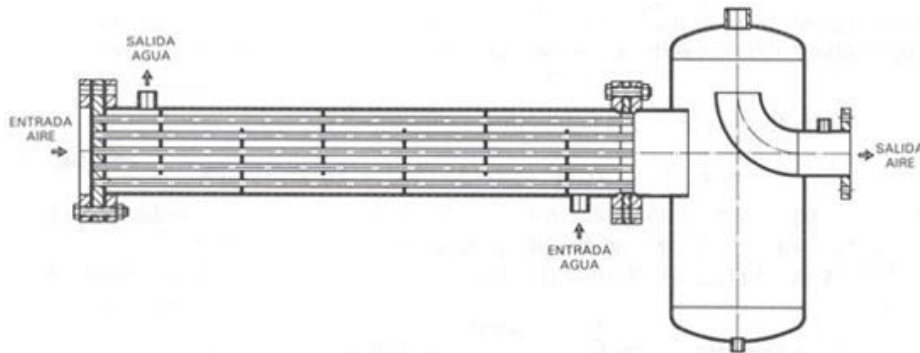
- ❑ After the compression, the air is passed through a cooling system to promote the water condensation.
- ❑ They are heat-exchangers, either air or water cooled.
- ❑ In the industrial coolers, the outlet temperature is often 25°C , in such way, the water is eliminated up to 70-80%.
- ❑ The **air cooler** consists of a series of pipes through which the compressed air passes and that externally receive another air current generated in a fan. They include a final water tank for condensed water.
- ❑ The outlet temperature of the compressed air is often 15°C over the cooling current generated in the fan.





3.4. Air treatment.

- ❑ The **water coolers** includes a series of pipes (tube arrays) that are inside another pipes, with higher diameter, through which there is a cooling water passing, in the opposite direction. They can be horizontal and vertical. They also include a small tank to drain the condensed water.
- ❑ It is guaranteed that the exit temperature of the air would be less than 10°C over the cooling water temperature.
- ❑ The calculation of these elements is done as in any other heat exchanger, considering the heat transmission as a function of the surfaces and the temperature difference.
- ❑ For a water cooling system, it must be kept in mind the head loss introduced in the system.





3.4. Air treatment.

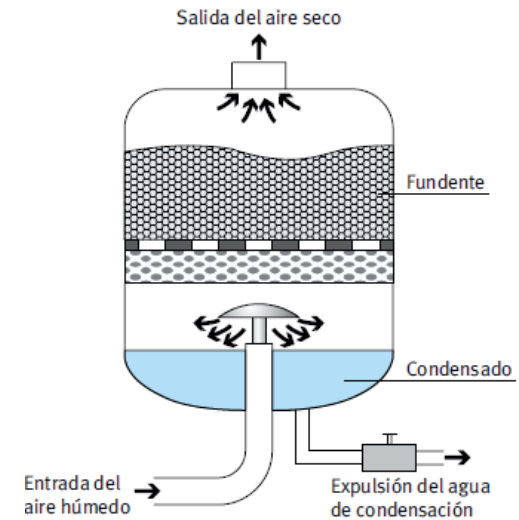
Dryers.

- ❑ Even if after-coolers reduce the temperature to reach 10-15°C over the cooling fluid, it is quite usual that the consumption (use) points could have even lower temperature (for instance, by night).
- ❑ To avoid condensations in the delivery pipes, additionally, some air **dryers**, to even reduce the air dew point.
- ❑ The lower the dew point is, the less humidity remains in the air.
- ❑ There are three main types of air dryers:
 - Absorption (coalescent)
 - Adsorption (drying effect)
 - By cooling (cold).



3.4. Air treatment.

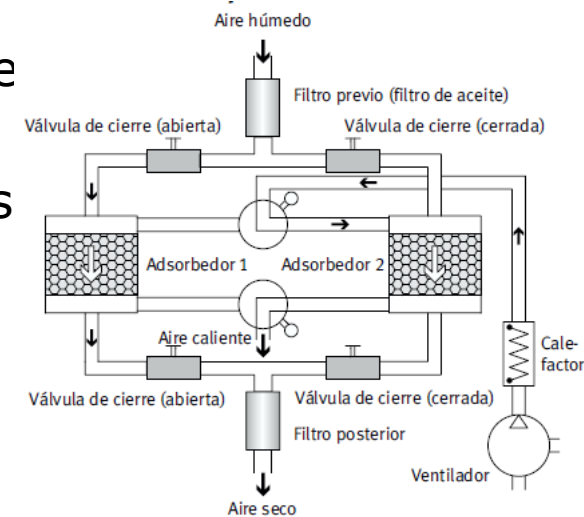
- ❑ The **absorption drying** it is a pure **chemical process**.
- ❑ The compressed air passes through a **drying material**, usually with Calcium chloride (dried gypsum, for instance) that reacts with the humidity to create a water based solution, that is afterwards eliminated.
- ❑ Advantages: this procedure is easy to implement, it does not have a mechanical wearing (no additional pieces) and no additional energy is needed.
- ❑ Disadvantages: the drying agent must be replaced regularly, the chemical additives are corrosive (a strong filtering process is needed) and the inlet temperature must be below 30°C.
- ❑ At relative pressure of 7 bar, it provides dew temperatures of around 5°C.





3.4. Air treatment.

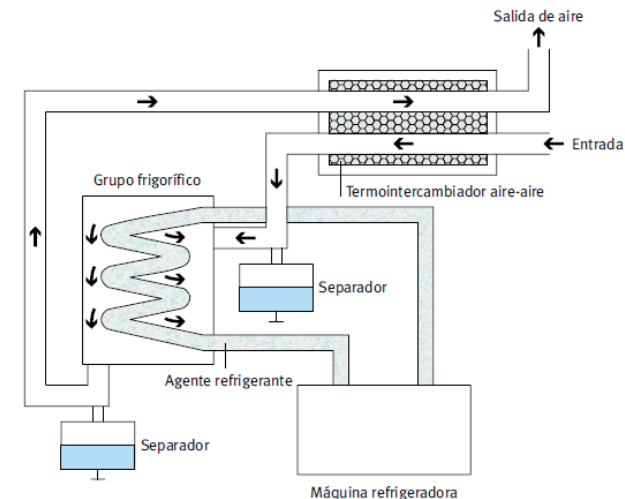
- ❑ The **adsorption drying** it is a **physical** procedure
- ❑ The drying material it is a grainy and might be considered as a porous media. This material keeps the water when the pressurized air is passed through. Often, it is a Silicon dioxide or activated alumina, often known as **gel**.
- ❑ The storage capacity is limited and the drying material must be replaced each 2/3 years.
- ❑ Externally, they are made with two twin tanks, both filled with adsorption material and that operate intermittently as in drying and regeneration sequences. The flow direction is controlled by a timing device in the directional valve.
- ❑ Usually, some heat is needed.
- ❑ Dew points extremely low are obtained, down to **-30°C**.





3.4. Air treatment.

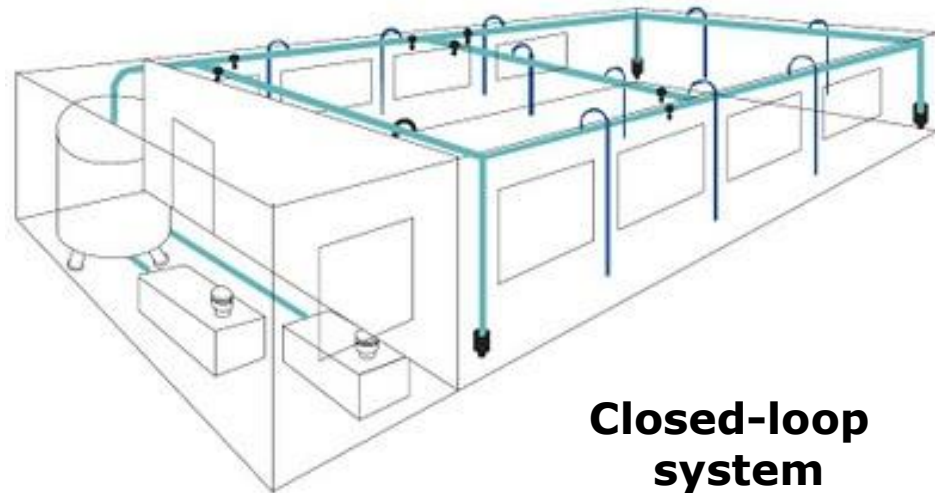
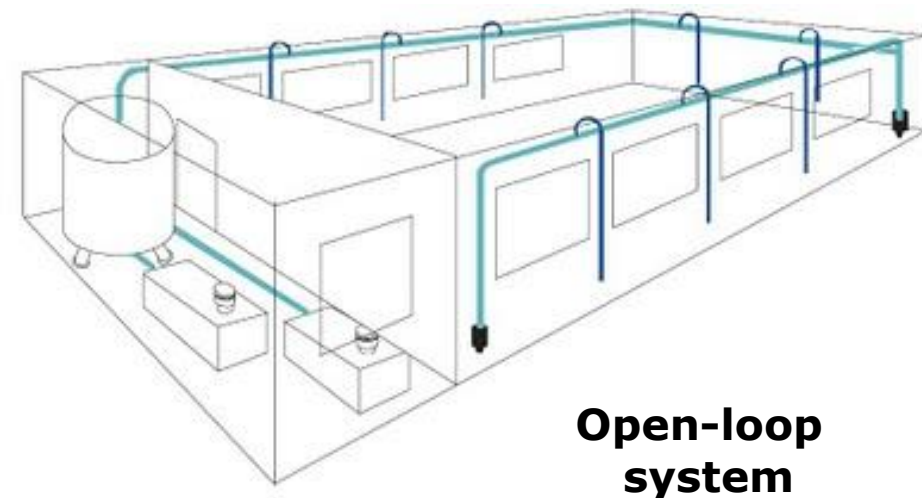
- ❑ The **drying by cooling** it is based on the temperature decrease of air to reach the dew temperature.
- ❑ The compressed air cooling is done using a fridge or refrigeration cycle. Through the evaporator coil the air interchanges heat with the a refrigerator fluid, often made with a mixture 25% glycol in water.
- ❑ Beforehand, a pre-cooling is done in another heat exchanger. An air/air cooling is used (two different exchangers are used).
- ❑ The air exit temperature is around 2 or 3°C.
- ❑ The cost of the drying process for compressed air could be around 10-20% of the generation costs.





3.5. Air distribution.

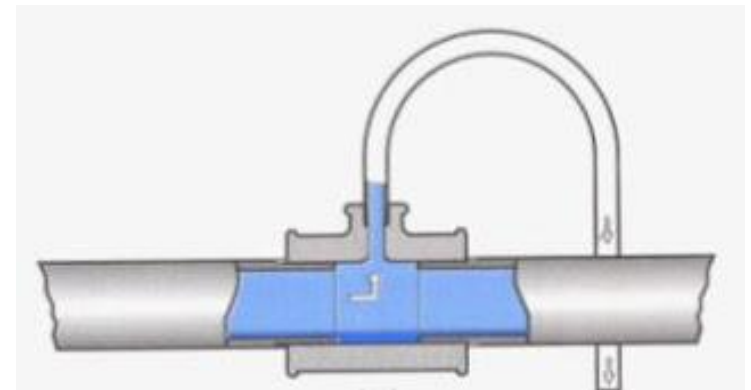
- ❑ The **open-loop system** is made by pipes that split to reach the final consumption points. The open-loop is the most economical solution, as it uses the less piping length.
- ❑ The **closed-loop system** it is closed at its end, making a ring-like system. For such layout, a optimum flow rate distribution is obtained and also continuum delivery when leakages or breaking up of any pipe, by the use of valves. On the other hand, it makes the head losses very low in all the system.





3.5. Air distribution.

- ❑ The pipes that constitute the system are placed in the upper part of the room or industrial unit, with supports every 3-4 meters.
- ❑ The pipes must have also a minimum number of elbows, placing (when possible) bends with a wide curvature radius.
- ❑ The pipes should also include a slope of 6-10 millimetres each meter (**0.6-1%**) and, if possible, never less than 3 mm in the flow direction.
- ❑ The secondary pipes should be connected in the upper part of the main pipe, to avoid the dust or particles transport to the final consumption point.
- ❑ Besides, it could be convenient to install automatic drains, where it can be good design criteria to change the pipes section there.



Piping connection



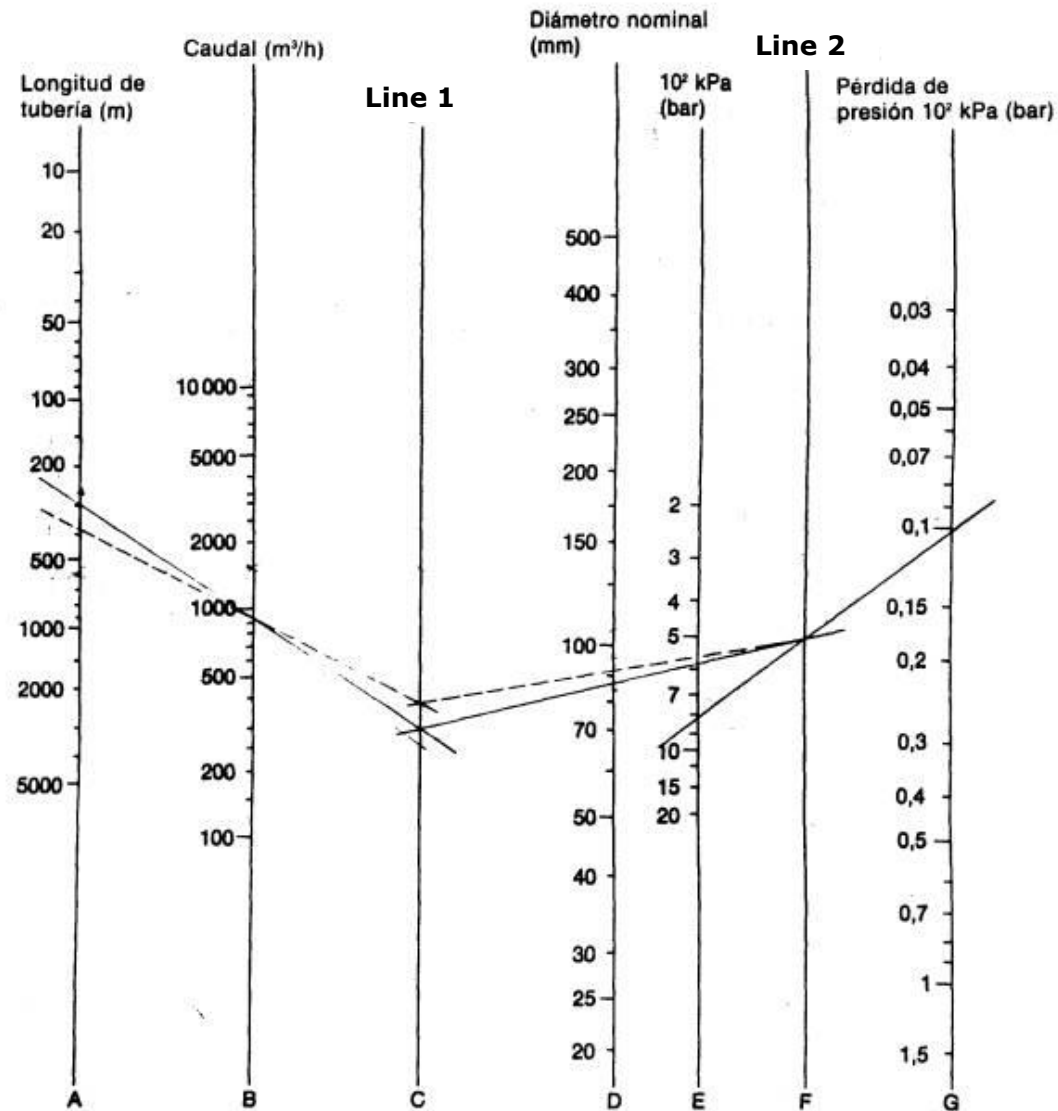
3.5. Air distribution.

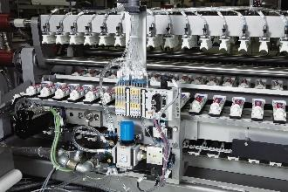
- ❑ The pipes cost is a quite important part of the initial system's cost.
- ❑ The sizing of the air pipes and secondary pipes it is well established considering the limit on the air speed, either with the head losses criteria or the maximum admissible speed:
 - Maximum head losses up to 2% of the working pressure. It is never recommended to reach the 0.3 bar in head losses (very seldom, values of 0.5 bar are admissible).
 - Maximum air speed of 6 m/s in the main pipes, that may reach up to 20 m/s in secondary pipes, near the consumption points.
- ❑ The pipe materials can be: steel or metal (standard gas pipes, SGP, the most common ones), stainless steel pipes (big diameters for long systems), cooper pipes or rubber pipes (flexible hoists) and PVC or nylon pipes, for inter-connection of the pneumatic elements.



3.5. Air distribution.

- ❑ For the quick calculation of the pipes diameters, **nomograms** are used.
- ❑ They allow a quick calculation of the pipe diameter needed to keep the head losses among a reasonable limit for a given flow rate at a known working pressure. In the graph, this is done with lines 1 and 2.
 - **Example:** full lines, 300 m and 900 m³/h lines, at 8 bar and maximum allowable pressure drop of 0.1 bar. The obtained diameter is around 85 mm.





3.5. Air distribution.

- ❑ The minor losses elements are also considered, usually as **equivalent length**.
- ❑ **Regulations** on gas pipes, according to standards (metal or other material).

Equivalent length (m)

D (mm)	Tipo de accesorio							
	Codo	Codo 90º	Curva 90º	Curva 180º	Válv. Esf.	Válv. Comp.	T – std	T – lat
15	0.3	1	0.1	0.5	0.8	0.1	0.1	0.5
25	0.5	1.6	0.2	0.8	1.4	0.2	0.2	0.9
30	0.7	1.8	0.4	1.1	2	0.3	0.4	1.4
40	0.8	2.2	0.5	1.2	2.4	0.3	0.4	1.6
50	1.1	2.6	0.6	1.7	3.4	0.4	0.5	2.1
65	1.4	3	0.8	2	4	0.5	0.7	2.7
80	1.8	3.9	0.9	2.6	5.2	0.6	0.9	3.7
100	2.4	5.4	1.2	3.7	7.3	0.9	1.2	4.1
125	3.2	7.1	1.5	4.1	9.4	1.2	1.5	6.4

**Standard pipe
diameters (mm)**

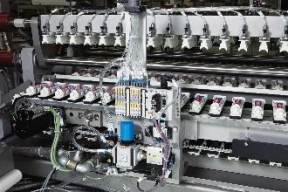
D (mm)	D (in)	Dext (mm)	e (mm)	m (kg/m)
6	1/8	10.35	2.00	0.419
8	1/4	13.8	2.30	0.652
10	3/8	17.3	2.30	0.851
15	1/2	21.7	2.80	1.310
20	3/4	27.2	2.80	1.680
25	1	34.0	3.20	2.430
32	1 1/4	42.7	3.50	3.380
40	1 1/2	48.6	3.50	3.890
50	2	60.3	3.65	5.100
65	2 1/2	76.1	3.65	6.510
75	3	88.9	4.05	8.470
100	4	114.3	4.50	12.100



3.5. Air distribution.

Delivery system for final use.

- ❑ Apart of the post-coolers, dryers and filters in the main line, it is necessary to clean the air as close as possible to the final **element or consumption point**.
- ❑ The treatment includes, as a first step the **filtering** of the particles: carbonized oil, small metal pieces from the pipes and other possible dust from the material wear (small oily matter pieces).
- ❑ Besides, it is not rare the need for a pressure control in the discharge so, very often, **pressure control valves** are used.
- ❑ Even though up to date it is not needed to lubricate the modern pneumatic components, it is often common to **lubricate**, seeding some oil into the compressed air.
- ❑ The elements that include filter, pressure control and lubricators are often combined into a service compact unit: **Units FCL** (Filter-Control-Lubrication).

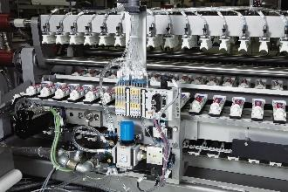


3.5. Air distribution.

Air quality.

□ The international standard **ISO 8573-1:2010** that grades the air quality according to the dust values (measuring the size of the solid suspended particles and their concentration), water (according to the dew point at pressure and the vapour content in the air) and oil (due to the total oil concentration in the air in the form of aerosols, liquids or vapours).

Class		Particles			Humidity and liquid water		Total oil ^[10]		
		0,1.. – 0,5 µm [1,6,9]	0,5.. – 1 µm [1,6,9]	1.. – 5 µm [1,6,9]					
0	↔	As specified and more stringent than class 1			As specified and more stringent than class 1		As specified and more stringent than class 1		
1	↔	≤ 20.000	≤ 400	≤ 10	≤ -70°C	[2,5]	≤ 0,01 mg/m ³	[1,3]	
2	↔	≤ 400.000	≤ 6.000	≤ 100	≤ -40°C		≤ 0,1 mg/m ³		
3	↔	---	≤ 90.000	≤ 1.000	≤ -20°C		≤ 1 mg/m ³	[1,4]	
4	↔	---	---	≤ 10.000	≤ +3°C		≤ 5 mg/m ³		
5	↔	---	---	≤ 100.000	≤ +7°C				
6	↔	≤ 5 mg/m ³		[1,7]	≤ +10°C	[1,8]			
7	↔	5.. – 10 mg/m ³			≤ 0,5 g/m ³				
8	↔				0,5.. – 5 g/m ³				
9	↔				5.. – 10 g/m ³				
X	↔	> 10 mg/m ³		[1,7]	> 10 g/m ³		> 5 mg/m ³	[1,4]	



3.5. Air distribution.

Air quality.

□ To design the air quality a value in the form of: **ISO 8573-1 X.Y.Z**, is to be used, where:

- **X** is the value to indicate the type of solid particles in the air, according to the table (previous slide).
- **Y** is the humidity value, according to the table.
- **Z** is the value that indicates the kind of oil, according to the table.

Example:

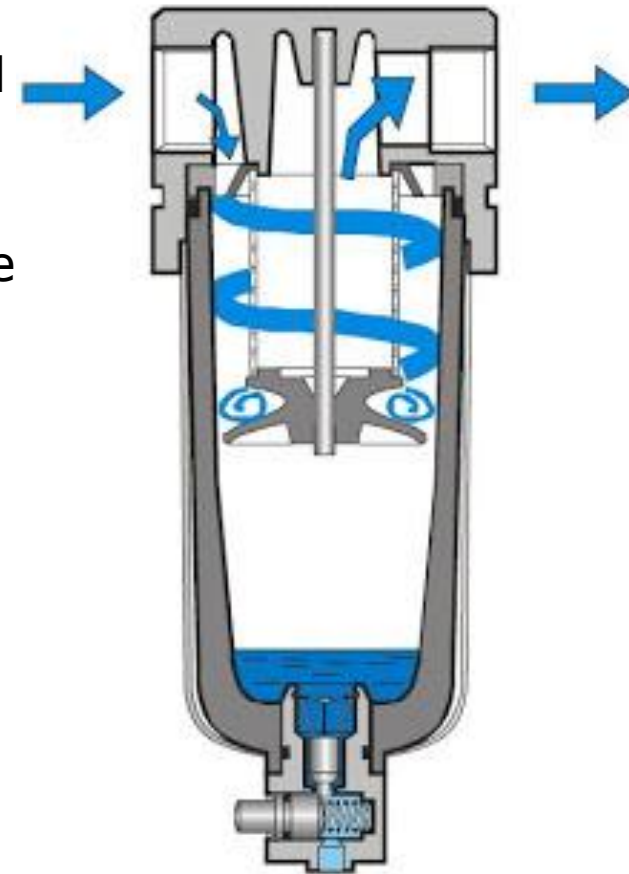
IG4 Aire para aplicaciones industriales: ISO 2.4.1	Eliminación eficiente de partículas sólidas y aceite. Se mantiene el punto de rocío a presión ISO Clase 4 o una Humedad Relativa (HR) del 30% (o inferior).	Fabricación general, troquelado de metales, herramientas neumáticas, estampación, montaje, pintura y acabado	
IG4 inodoro Aire para aplicaciones industriales: ISO 2.4.1 inodoro	Eliminación eficiente de partículas sólidas y aceite y vapor de aceite. Se mantiene el punto de rocío a presión ISO Clase 4 o una Humedad Relativa (HR) del 30% (o inferior).	Alimentos y bebidas, mezcla de materias primas	
IG6 Aire para aplicaciones industriales: ISO 2.6.1	Eliminación eficiente de partículas sólidas y aceite. Se mantiene el punto de rocío a presión ISO Clase 6 o una Humedad Relativa (HR) del 50% (o inferior).	Aspersión de arena, uso doméstico, construcción	



3.5. Air distribution.

- ❑ It has as main goal to remove the suspended particles of the air, the solid particles coming from the inlet or by the metal pieces being dragged by the air in the system and the liquid particles as water or oil, that may be in the system.
- ❑ The usual filter is consist of a bowl in which the air goes in and passes along a deflector plate with inclined slots. This plate derives the air and promotes a centrifugal effect. The solid particles are separated when they impact with the side walls of the bowl and fall down to the lower part. Then, the air is passed through a filter (surface filter) with a porosity between 5 and 45 microns, depending on the filtering degree required.
- ❑ Typically, they are cartridge units.

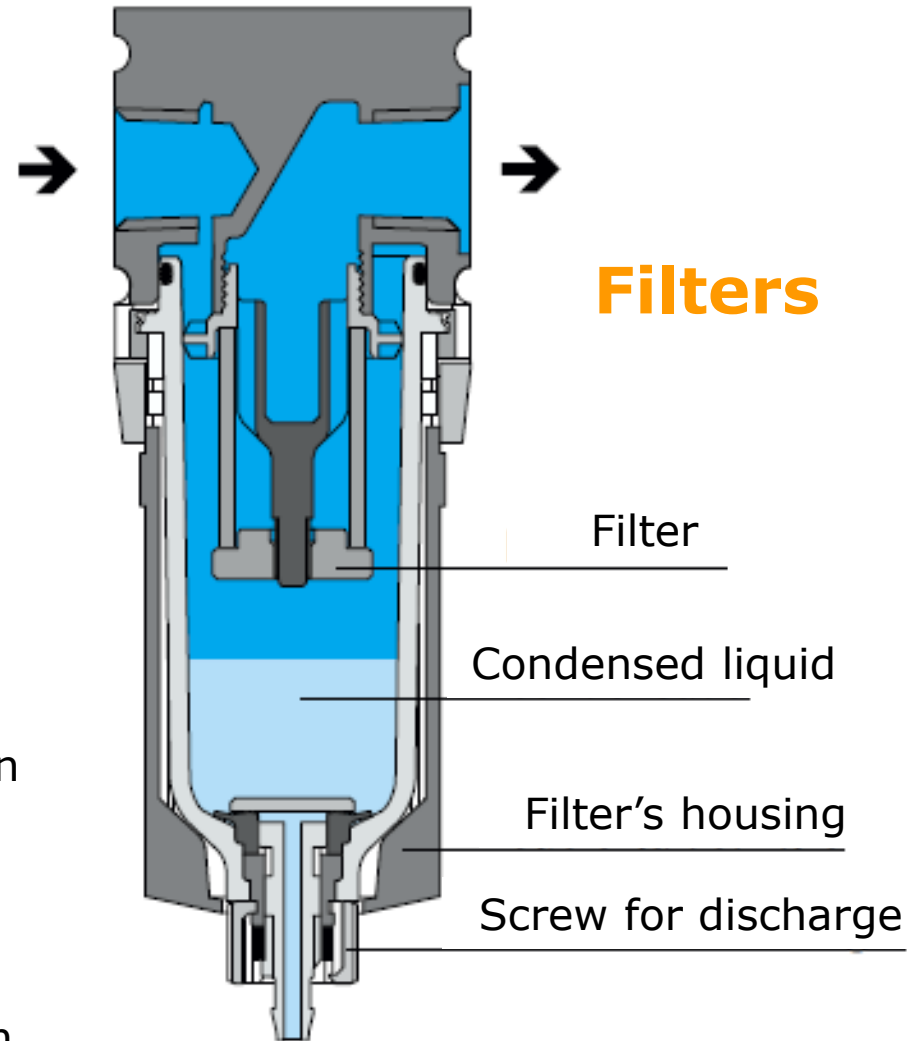
Filters





3.5. Air distribution.

- ❑ The centrifugal effect allows to condensate part of the water that may be carried out by the air. Therefore, at low speeds, they are not so effective.
- ❑ Filtering degree:
 - Standard filter: 5-40 microns.
 - Low filter: 1 micron.
 - Very low filter: 0.01 microns.
- ❑ The filtering size depends on two factors:
 - The maximum air flow rate used in the component.
 - The maximum pressure drop admissible in the current application (often, the manufactures, provide information graphs).

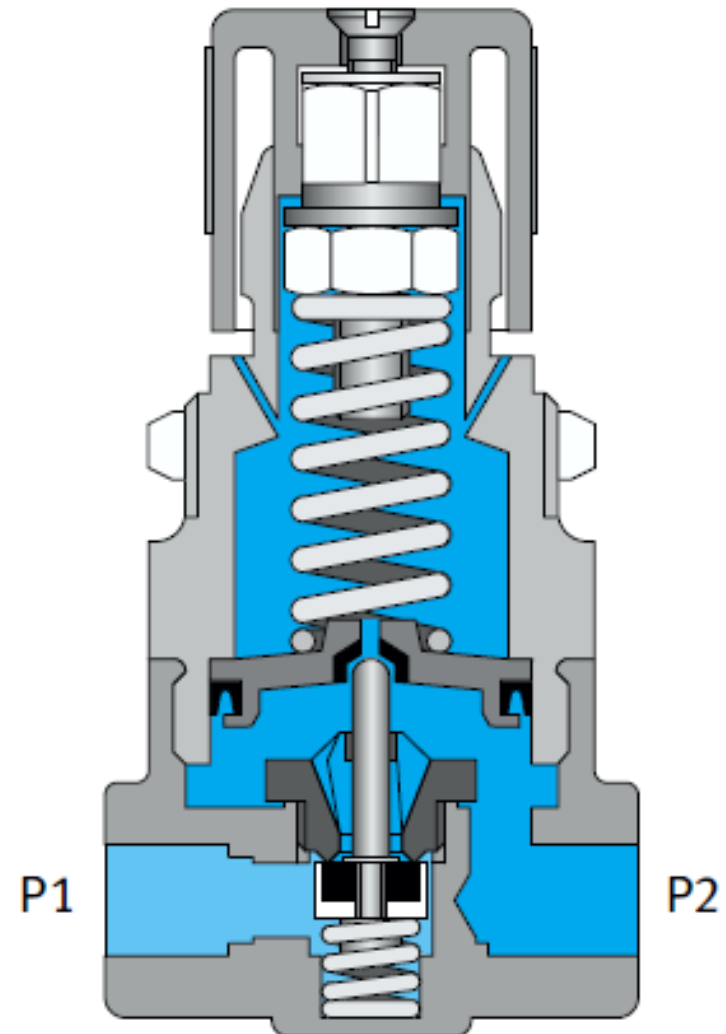




3.5. Air distribution.

Pressure control.

- ❑ The pressure control valves have a piston or diaphragm to compensate the outlet pressure against the force of the spring (externally operated).
- ❑ Its mission is to keep the outlet pressure P_2 as constant as possible, regardless the changes in the inlet pressure, P_1 .
- ❑ When increasing the working pressure, the compressed air may find a way out through an orifice.
- ❑ They can be internally piloted, obtaining a better precision and stability.

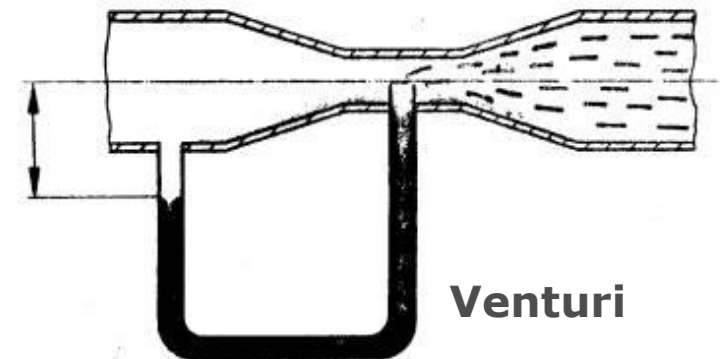
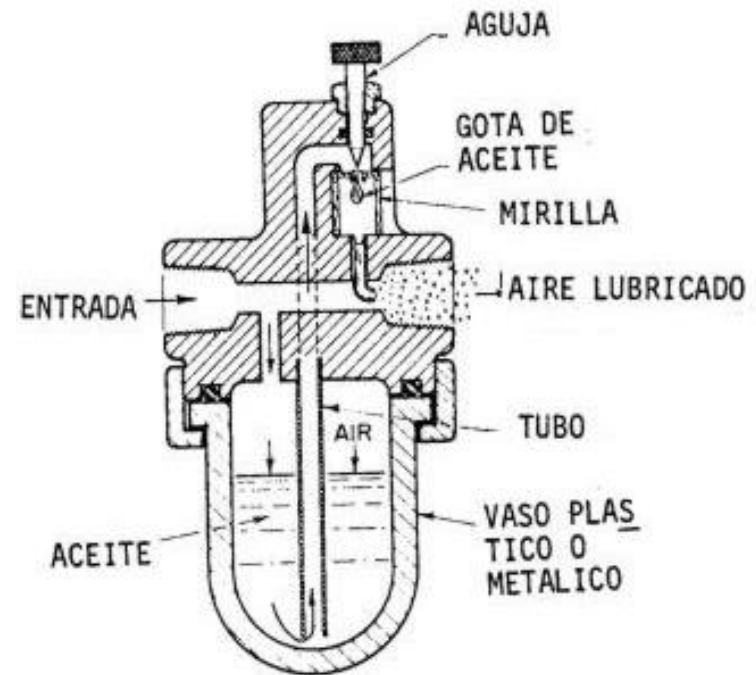




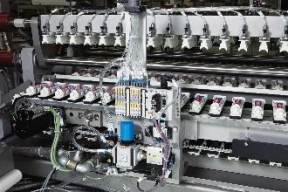
3.5. Air distribution.

Lubricators.

- ❑ With the lubrication, the wear is reduced and the friction losses and the protection against corrosion it is increased.
- ❑ It works as a **Venturi**. The lubricator gives, by the use of a restriction for the air, a pressure drop. In the droplet chamber, an aspirating effect is similarly produced, promoting the oil mixture in the air current. A cloud-spray effect is produced and then, the oil is transported by the air. The amount of oil in the air is controlled with a needle.



Venturi



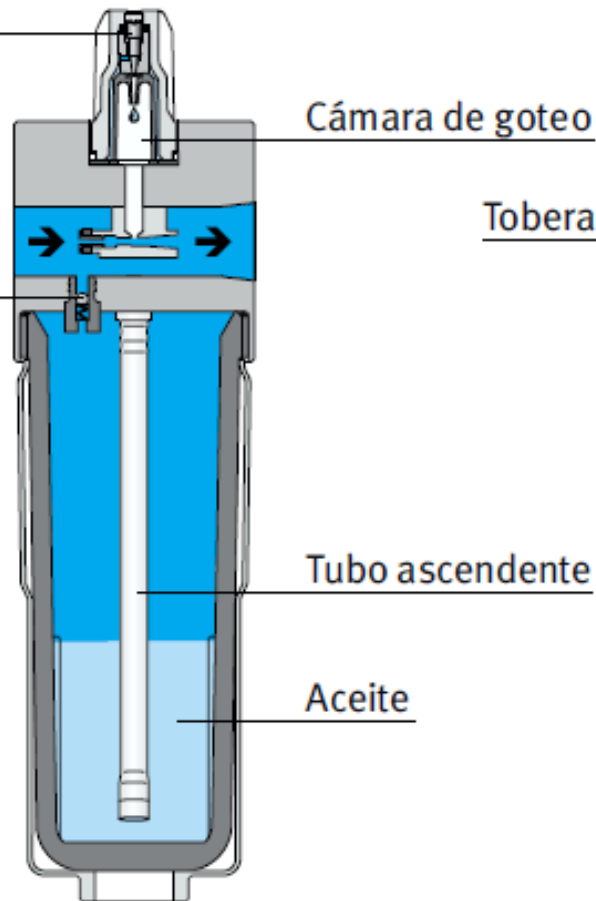
3.5. Air distribution.

Lubricators

Compressed air lubricators

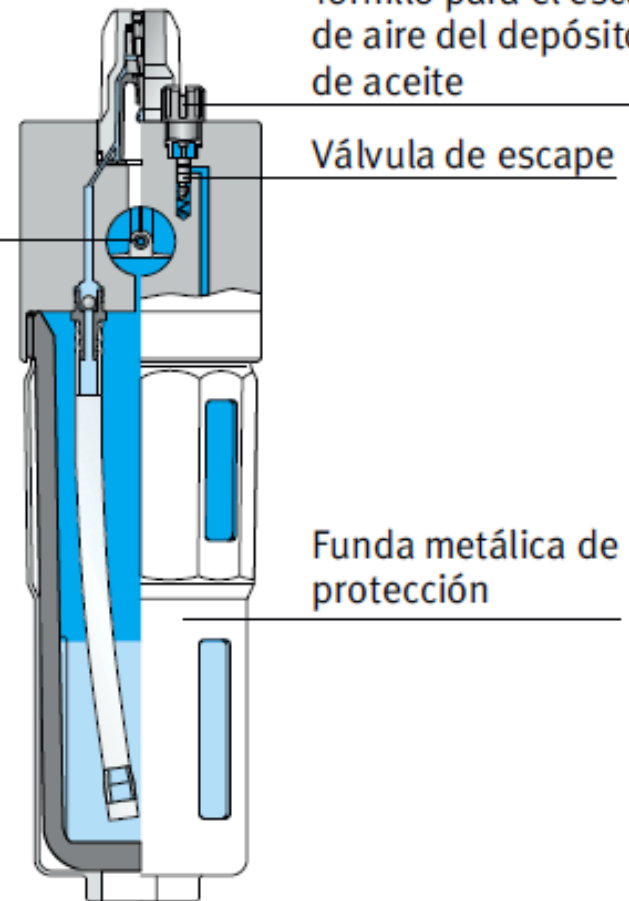
Tornillo de regulación
para la alimentación
de aceite

Válvula de
retención



Tornillo para el escape
de aire del depósito
de aceite

Válvula de escape

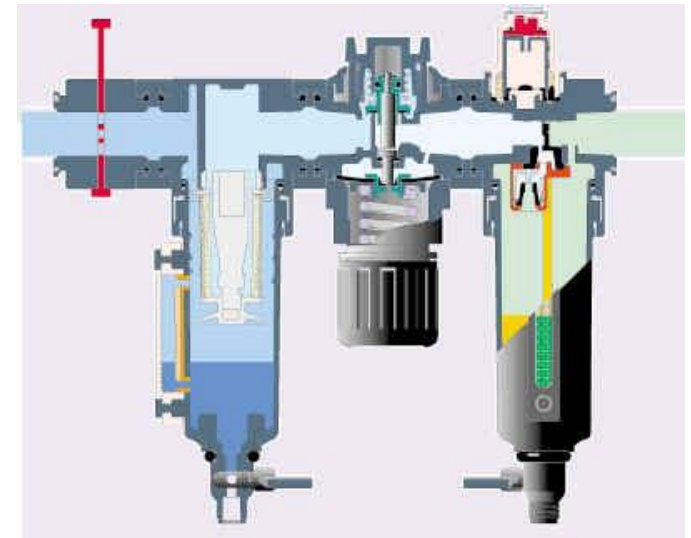


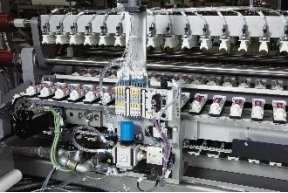


3.5. Air distribution.

FCL systems.

- ❑ The maintenance unit is made of a filter (F), the control (C) and the lubricator (L), in a modular configuration.
- ❑ This maintenance unit, that combines the functions of the three elements, it is normally placed at the air inlet of the consumption point.
- ❑ The maintenance unit should always be mounted with the elements in the established order, following the air flow direction.
- ❑ Its size must be chosen according to the maximum flow rate in the system (usually, the manufacturers give this information).





3.5. Air distribution.

Typical air consumption:

- ❑ Other consumption, besides the required in **pneumatic cylinders** (section by stroke).
- ❑ **Pneumatic tools:** blow guns, grinders, grease guns, air hammers, drills, riveters, carving tools, air bushing...
- ❑ A consumption according to the **load factors** (not all the tools working at the same time) is obtained.

TABLE C15.9 General Air Requirements for Tools

Tools or equipment	Size or type ^a	Air pressure, psi	Air consumed, scfm ^b
Hoists	1 ton	70–100	1
Blow guns		70–90	3
Bus or truck lifts	14,000-lb cap	70–90	10
Car lifts	8,000-lb cap	70–90	6
Car rockers		70–90	6
Drills, rotary	1/4-in cap	70–90	20–90
Engine, cleaning		70–90	5
Grease guns		70–90	4
Grinders	8"-in wheel	70–90	50
Grinders	6"-in wheel	70–90	20
Paint sprayers	Production gun	40–70	20
Spring oilers		40–70	4
Paint sprayers	Small hand	70–90	2–7
Riveters	Small to large	70–90	10–35
Drills, piston	1/2-in cap, 3-in cap	70–90	50–110
Spark plug cleaners	Reach 36–45	70–90	5
Carving tools		70–90	10–15
Rotary sanders		70–90	50
Rotary sanders		70–90	30
Tire changers		70–90	1
Tire inflaters		70–90	1 1/2
Tire spreaders		70–90	1
Valve grinders		70–90	2
Air hammers	Light to heavy	70–90	30–40
Sand hammers		70–90	25–40
Nut setters and runners	1/4-in cap to 3/4-in cap	70–90	20–30
Impact wrenches/screwdrivers	Small to large	70–90	4–10
Air bushings	Small to large	80–90	4–10
Pneumatic doors		40–90	2
File and burr tools		70–90	20
Wood borers	1–2 in	70–90	40–80
Rim strippers		100–120	6
Body polishers		70–90	2
Vacuum cleaners		100–120	6
Carbon removers		70–100	3
Sand blasters	Wide variation	90	6–400

^a 1 inch = 25.4 mm

^b 1 cubic foot = 0.0283 m³



3.5. Air distribution.

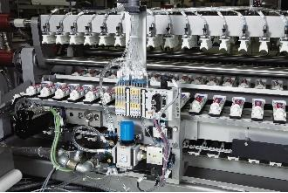
Airflow Considerations:

- ❑ There is no universal way to provide data on compressed air consumption or generation capacity. Some of the units most used in practice are:
 - ❑ **NL/min**: normal liters per minute. Flow at a certain point converted to reference conditions of **P=1.013 bar(a)**, **T=0°C** and **0%** relative humidity. They can also be at 15° C or 20° C.
 - ❑ **ANR**: (Reference Normal Atmosphere). Flow at a given point converted to some reference conditions of **P=1.013 bar(a)**, **T=20° C** and **hr=65%**.
 - ❑ **Scfm**: standard cubic feet per minute. Flow at a given point converted to some reference conditions of **P=14.7 psi(a)** (1.013 bar), **T=60° F** (~15° C) and **0%** humidity relative. They can also be found at other temperatures and relative humidities.
 - ❑ **FAD**: (Free Air Delivery, L/min). Flow at a certain point converted to the reference conditions established in the **ISO 1217** standard: **P=1 bar(a)**, **T=20° C** and **0%** relative humidity.
- ❑ For example, in **compressors** the value of the compressed air flow at the outlet of the refrigerator (1) is usually given, converted to FAD (0):

$$V_0 = V_1 \cdot \frac{T_0}{T_1} \cdot \frac{P_1}{P_{0c}}$$

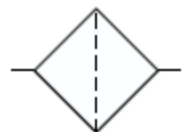
- Where P_{0c} is the absolute pressure at the inlet, corrected by the relative humidity value (although this correction is usually of little importance):

$$P_{0c} = P_o - P_v \cdot h_r \quad \mathbf{P_v}: \text{vapor pressure and } \mathbf{h_r}: \text{relative humidity}$$

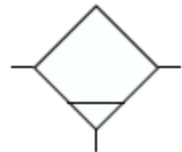


3.5. Air distribution.

Maintenance:



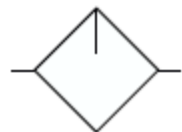
Filter.



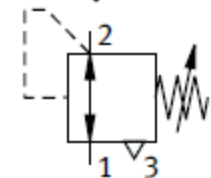
Water drain with manual operation.



Water drain with automatic operation.



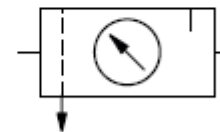
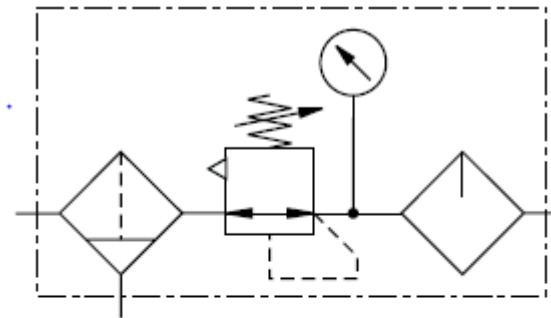
Lubricator.



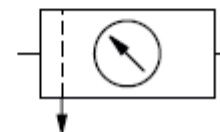
Pressure relief valve with variable discharge port.

Combined symbols:

FCL system (maintenance): Filter with water drain, pressure relief valve, manometer and lubricator.



Simplified representation (symbol)



Symbol without lubrication